



The Young and the Wild: What happens to Protoclusters forming at $z \approx 4$?

Rhea-Silvia Remus

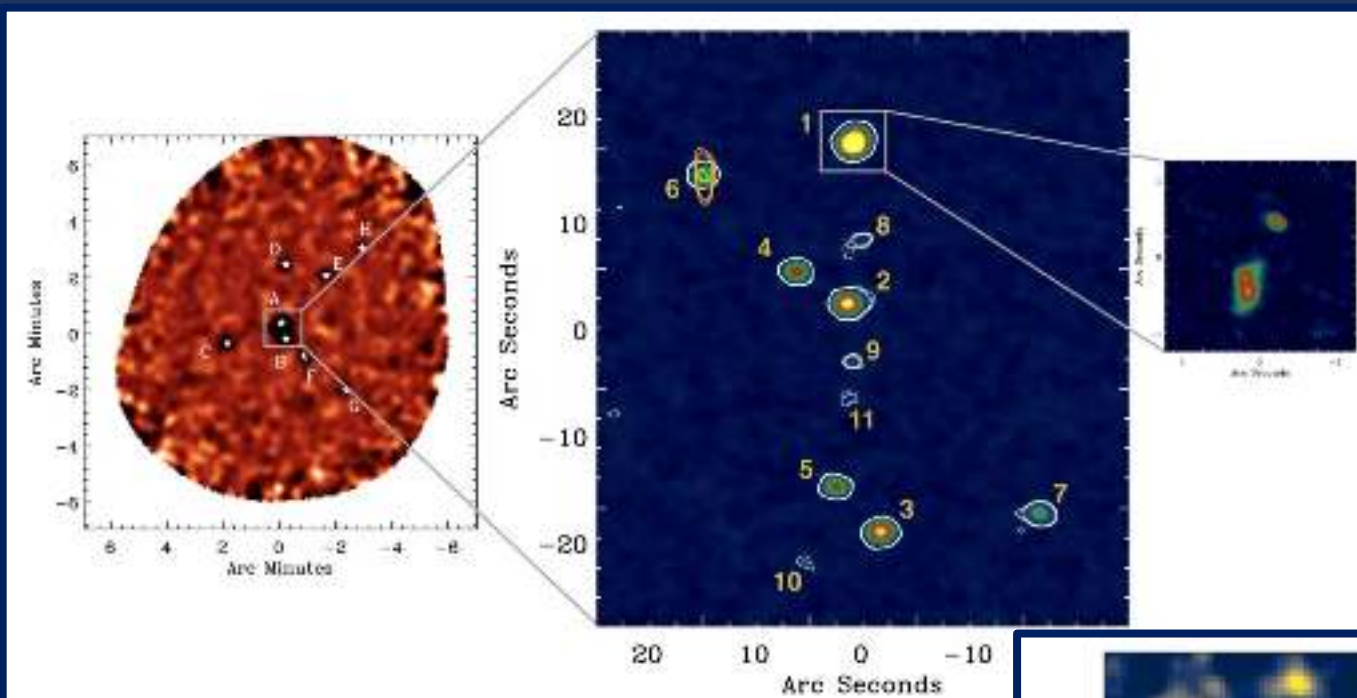
with Klaus Dolag, Felix Schulze, Elena Hernandez
and Scott C. Chapman, Helmut Dannerbauer



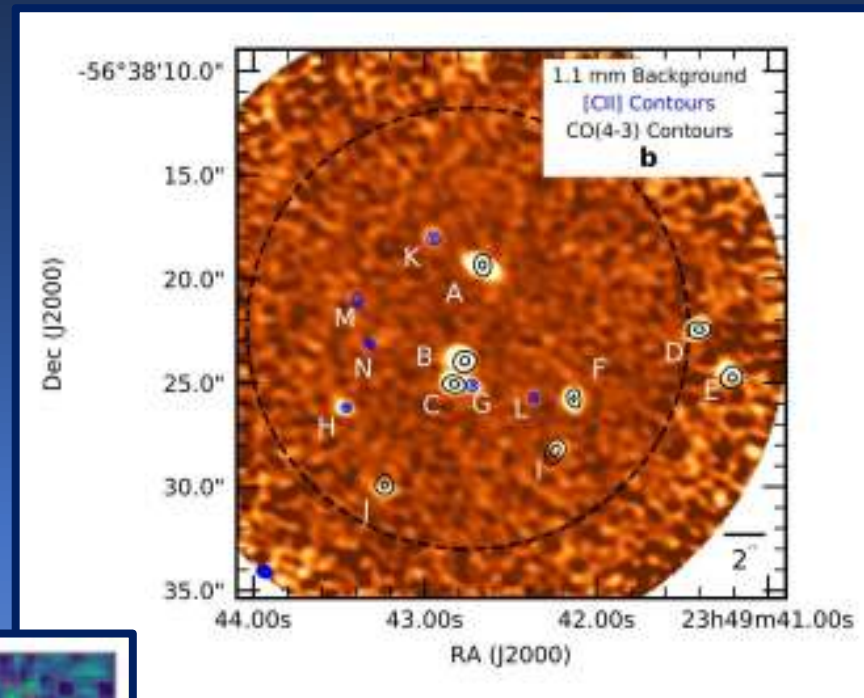
Protoclusters: Galaxies in Confinement, 03.09.2020



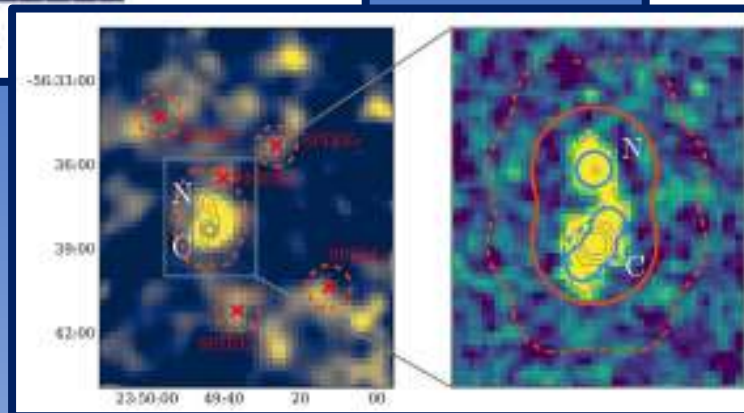
Protoclusters @ $z \approx 4.0$



Estimated $M_{tot} = 4.4 \times 10^{13} M_{\odot}$
 Oteo et al., 2018



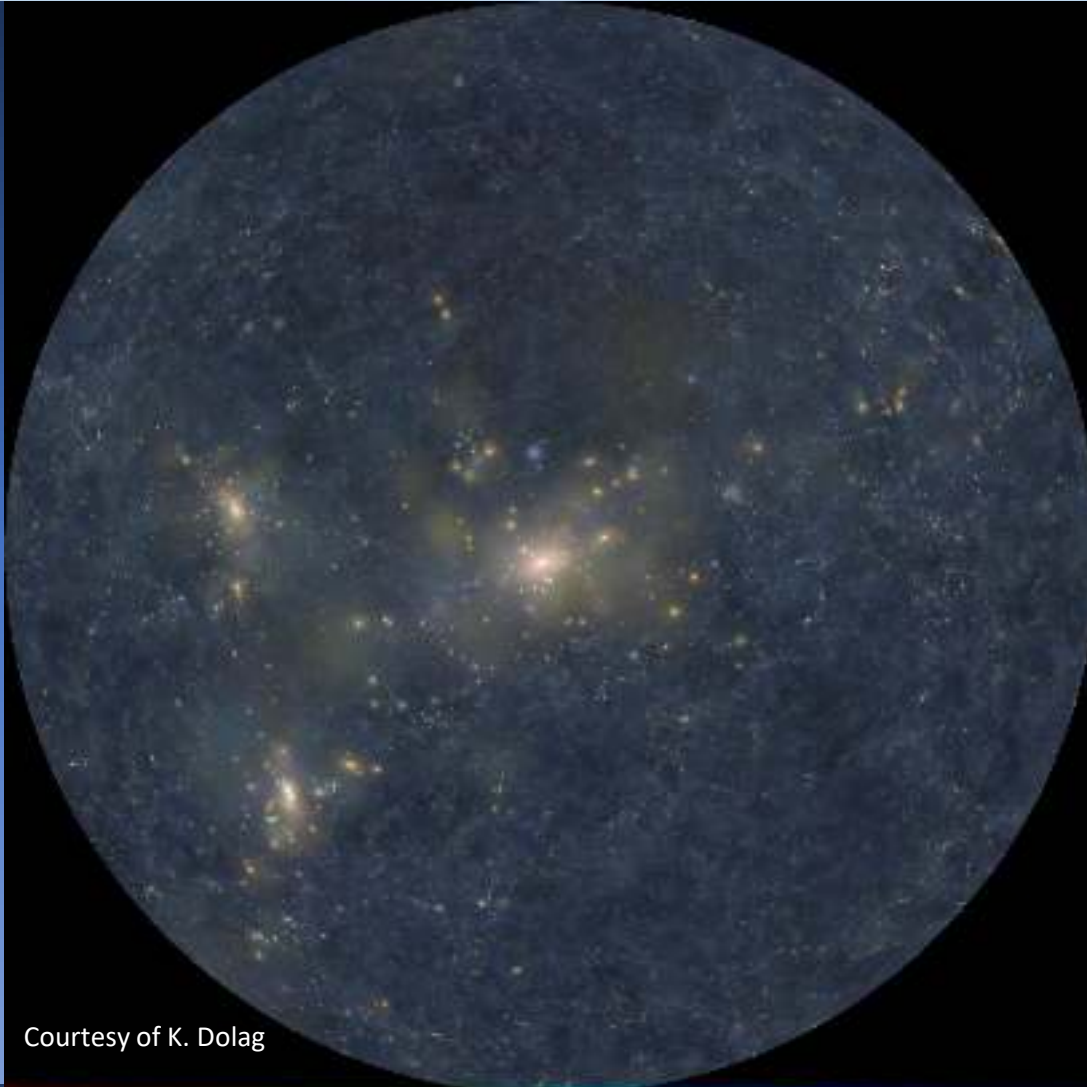
Estimated $M_{tot} = 1.16 \times 10^{13} M_{\odot}$
 Miller et al., 2018



Estimated $M_{tot} = 9 \pm 5 \times 10^{12} M_{\odot}$
 Hill et al., 2020

The Simulations: Magneticum

www.magneticum.org



Courtesy of K. Dolag

Box	Mpc/h	mr	hr	uhr
0	2688	y		
1	896	y		
2b	640	y	y	
3	128	y	y	(z=2)
4	48	y	y	y

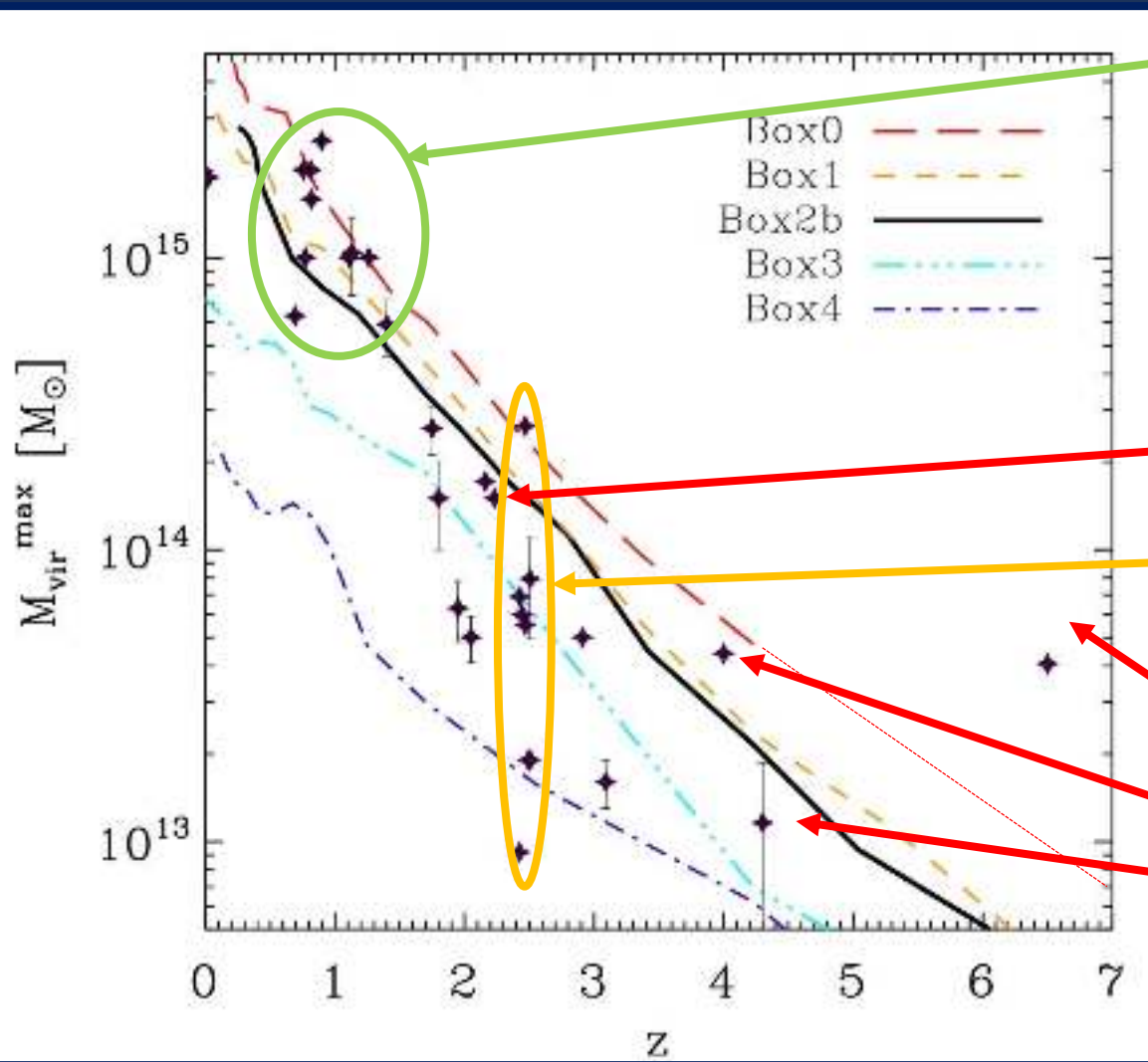
	mr	hr	uhr
$m_{DM} (M_{\odot}/h)$	$1.3 \cdot 10^{10}$	$6.9 \cdot 10^8$	$3.7 \cdot 10^7$
$m_{Gas} (M_{\odot}/h)$	$2.6 \cdot 10^9$	$1.4 \cdot 10^7$	$7.3 \cdot 10^6$

One gas particle can spawn up to 4 stellar particles.

- Modified SPH version of GADGET-3 (incl. thermal conduction)
- Feedback from stellar winds
- Feedback from AGN
- Metal enrichment and star formation follow pattern of metal production from SNIa, SNII & AGB
- Gas cooling depends on local metallicity



Simulations: Overview



ORELSE

(Tomczak et al., 2017)

Darvish et al., 2020

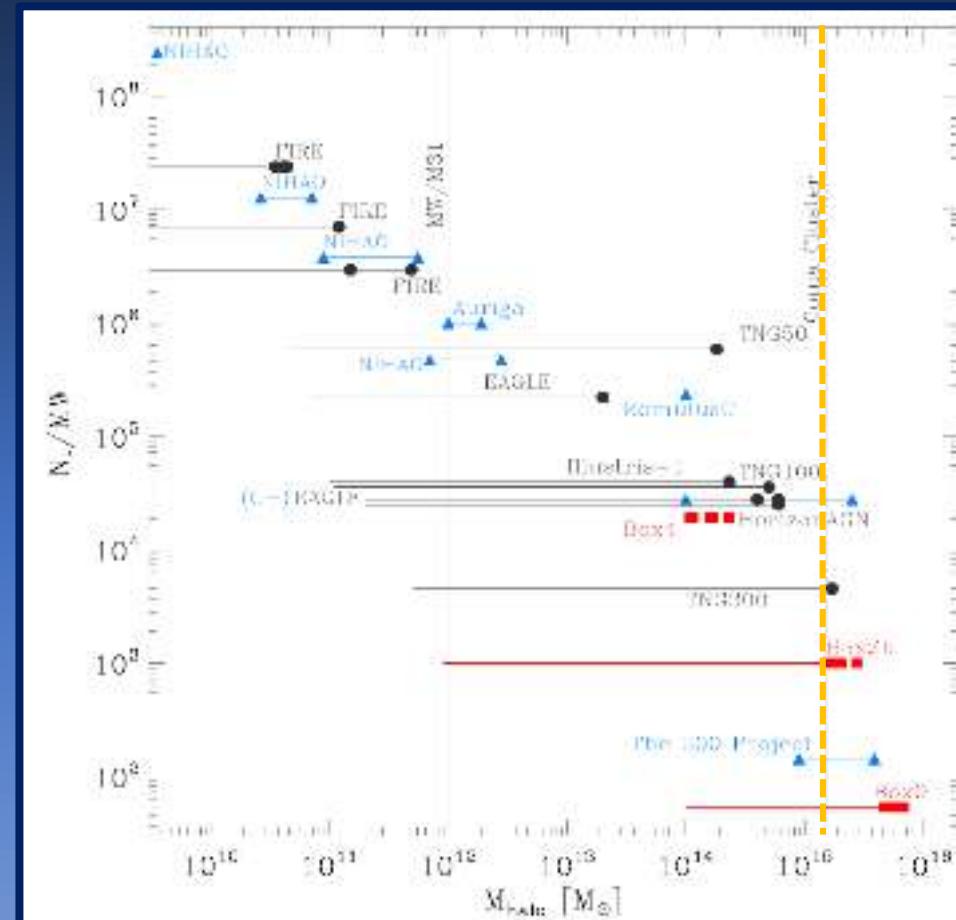
Hyperion

(Cucciati et al., 2018)

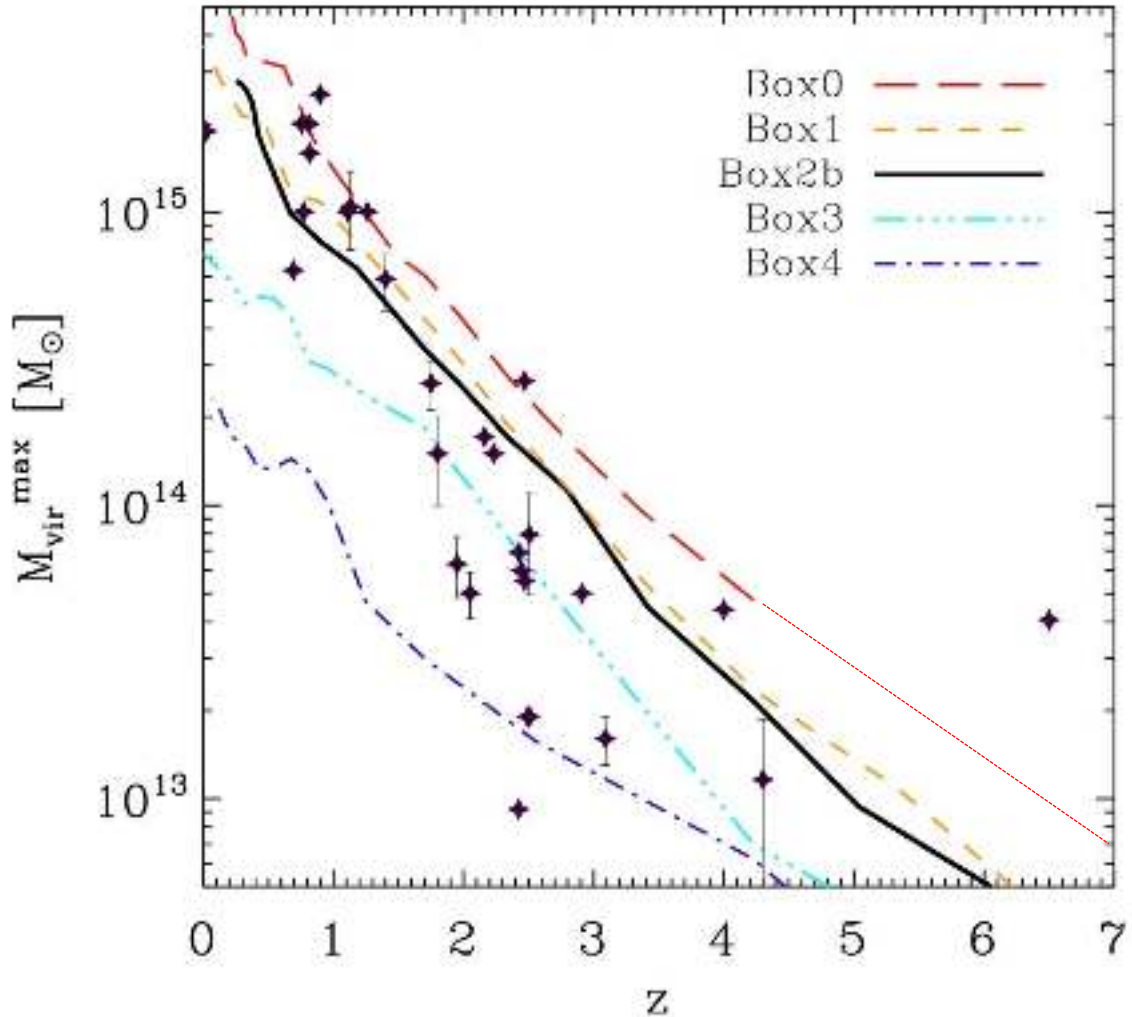
Chanchaiworawit
et al., 2019

Oteo et al., 2018

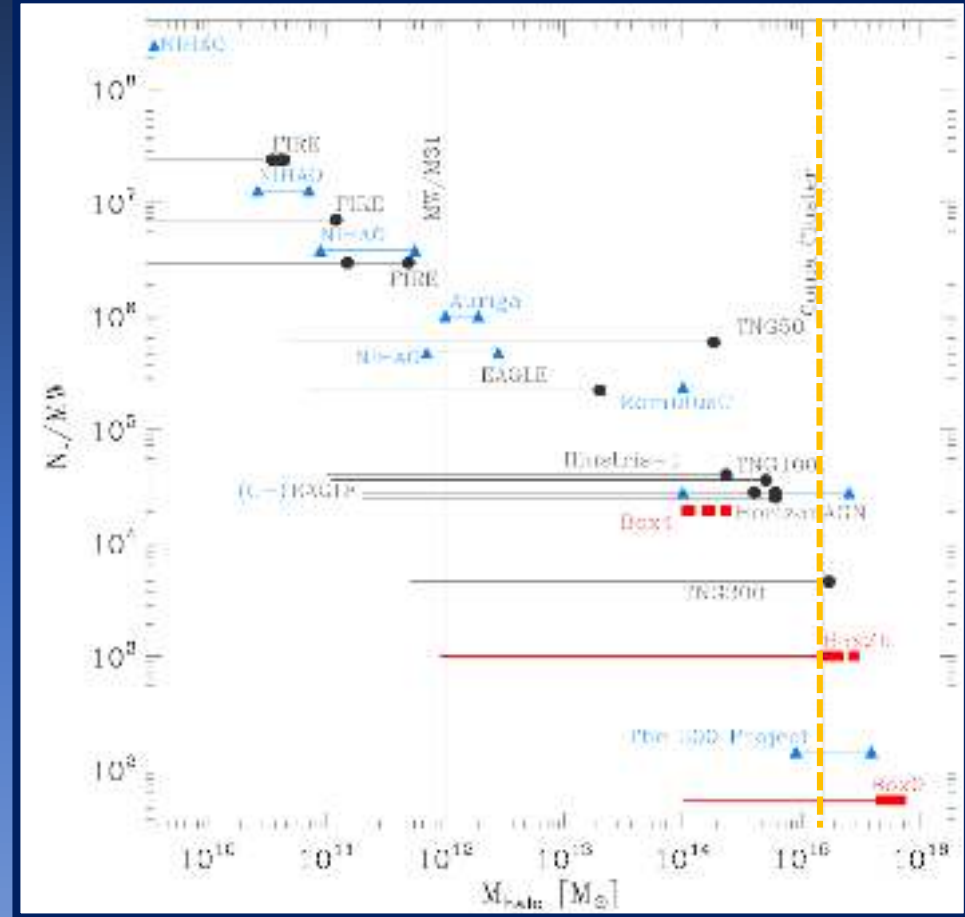
Miller et al., 2018



Simulations: Overview

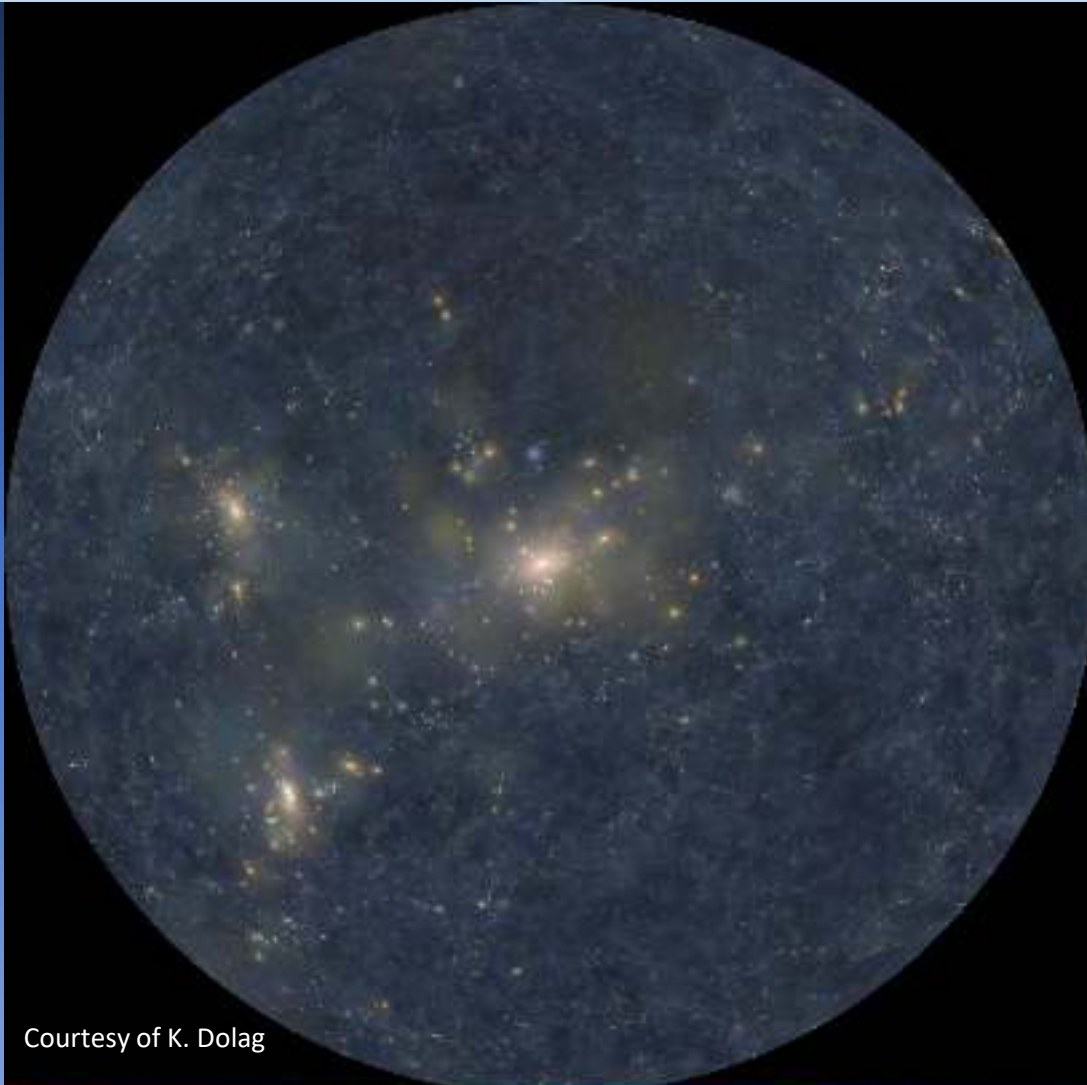


Box	Mpc/h
0	2688
1	896
2b	640
3	128
4	48



The Simulations: Magneticum

www.magneticum.org



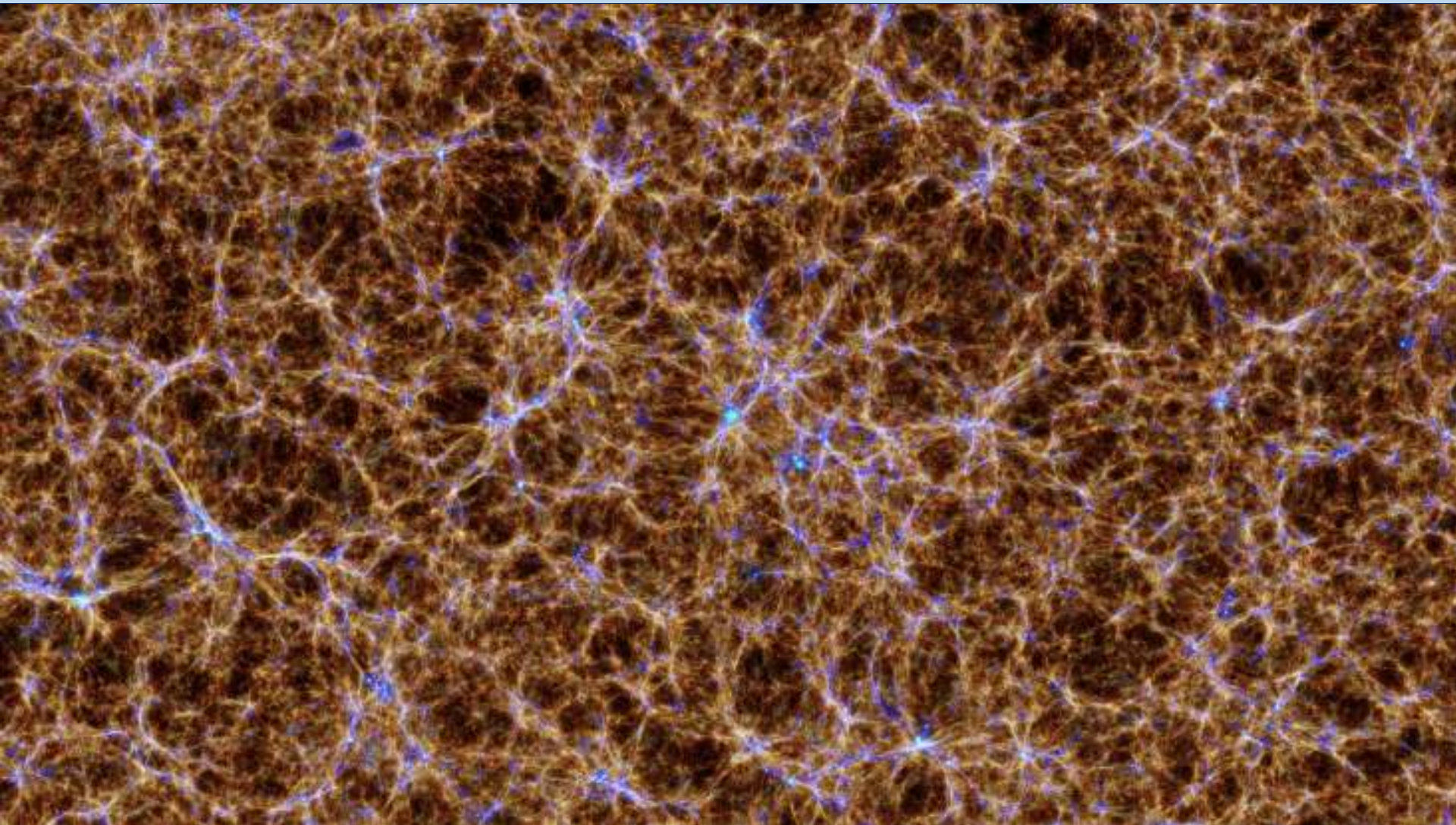
Courtesy of K. Dolag

Box	Mpc/h	mr	hr	uhr
0	2688	y		
1	896	y		
2b	640	y	y	
3	128	y	y	(z=2)
4	48	y	y	y

	mr	hr	uhr
$m_{DM} (M_{\odot}/h)$	$1.3 \cdot 10^{10}$	$6.9 \cdot 10^8$	$3.7 \cdot 10^7$
$m_{Gas} (M_{\odot}/h)$	$2.6 \cdot 10^9$	$1.4 \cdot 10^7$	$7.3 \cdot 10^6$

One gas particle can spawn up to 4 stellar particles.

- Modified SPH version of GADGET-3 (incl. thermal conduction)
- Feedback from stellar winds
- Feedback from AGN
- Metal enrichment and star formation follow pattern of metal production from SNIa, SNII & AGB
- Gas cooling depends on local metallicity



Magneticum Box2b-Simulation:

640 Mpc/h,

$$m_{DM} (M_{\odot}/h) = 6.9 \cdot 10^8$$

Millenium-Simulation:

500 Mpc/h,

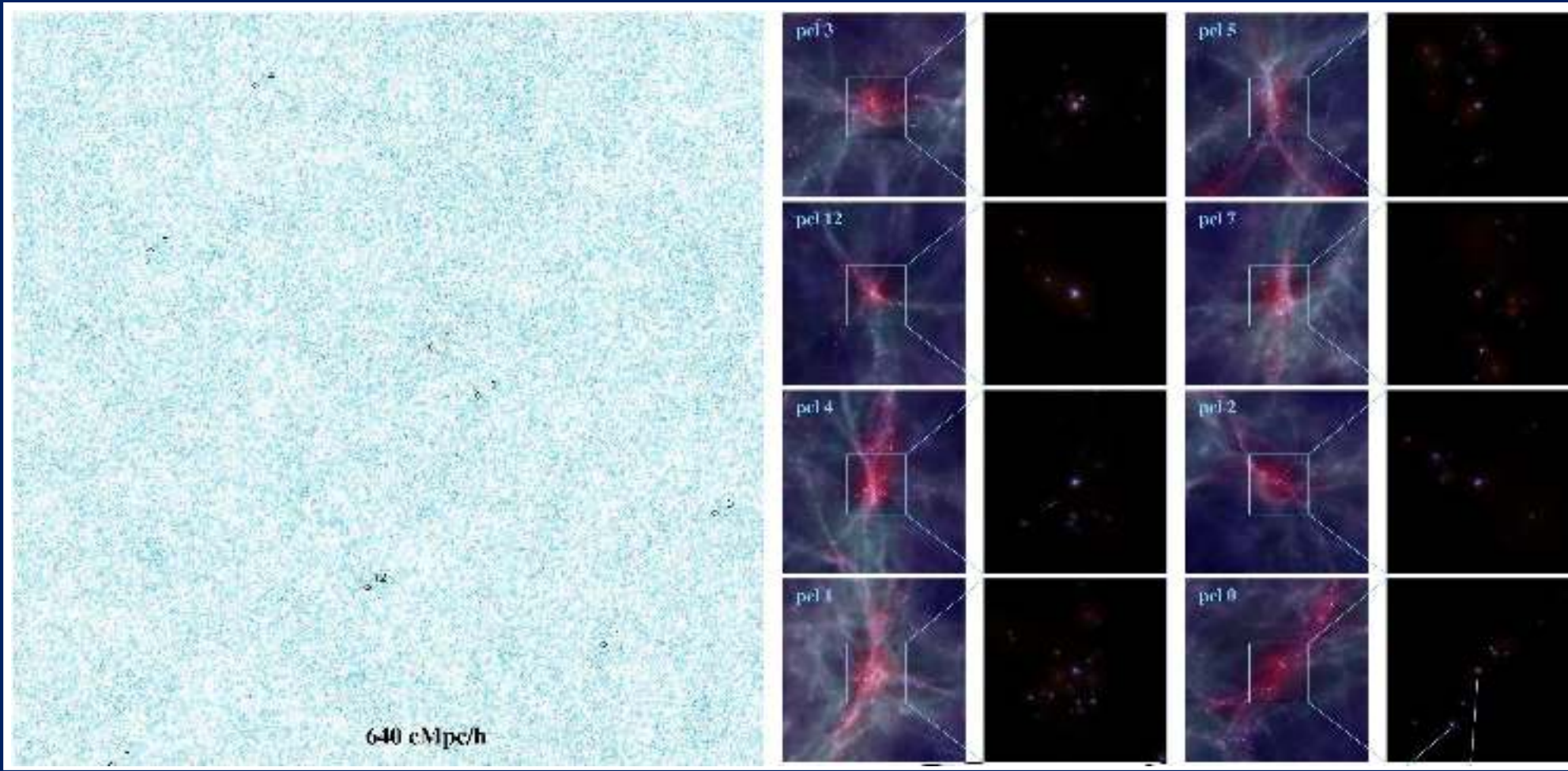
$$m_{DM} (M_{\odot}/h) = 8 \cdot 10^8$$

TNG300:

300 Mpc/h,

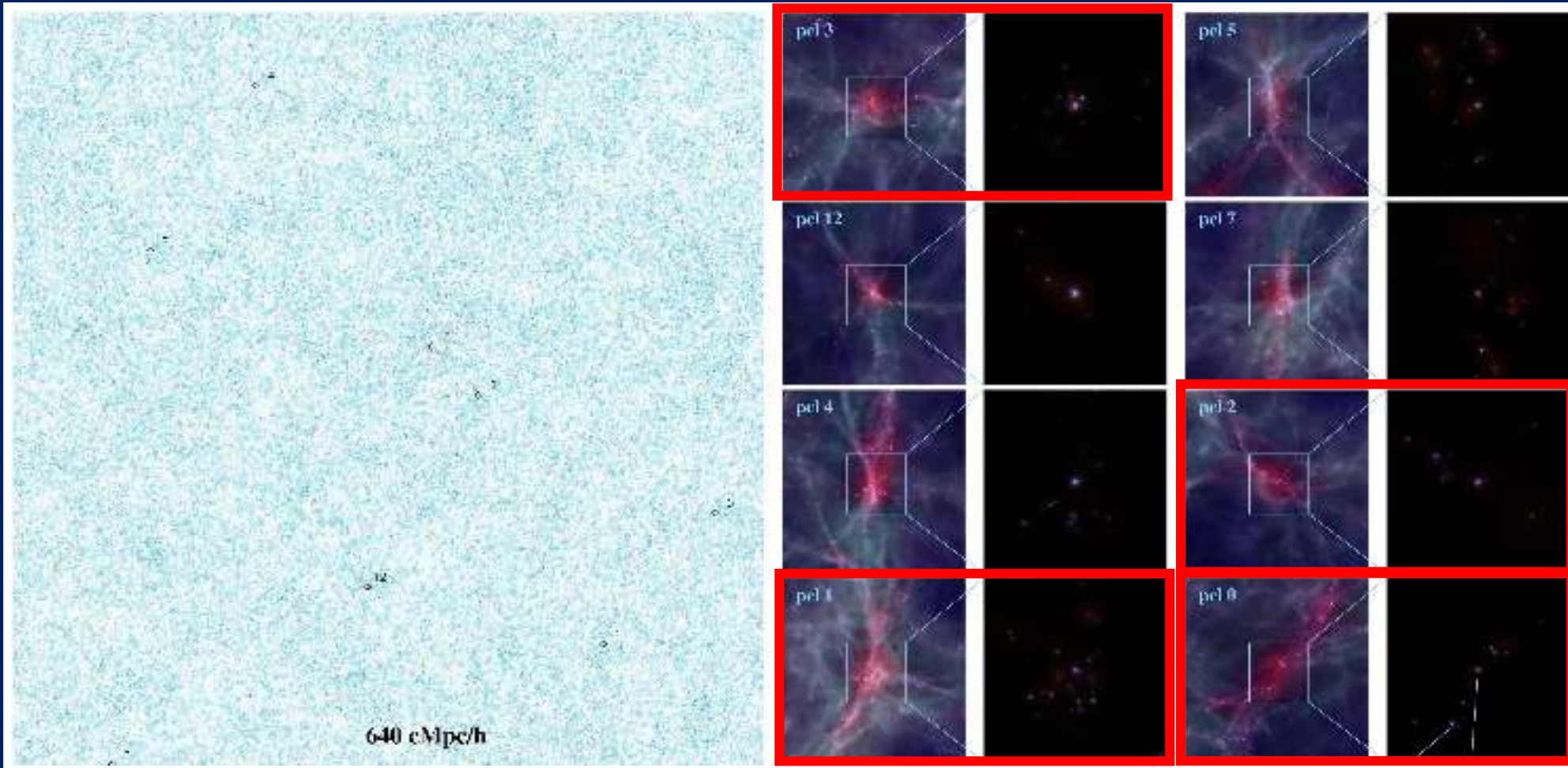
$$m_{DM} (M_{\odot}/h) = 6 \cdot 10^7$$

Protocluster in Magneticum



42 potential
Protocluster with
 $M_{vir} > 10^{13} M_{\odot}$

Protocluster in Magneticum

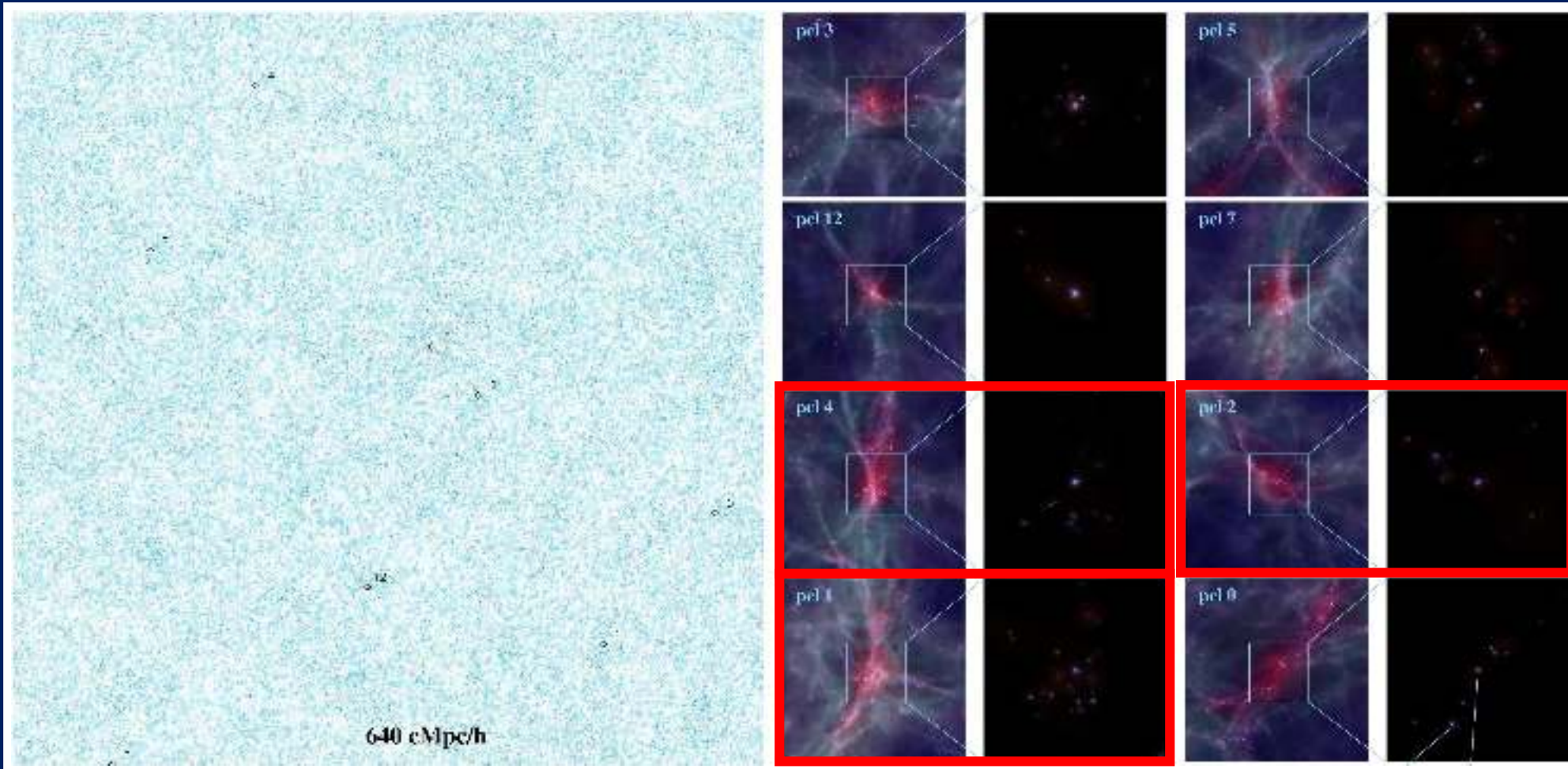


42 potential
Protocluster with
 $M_{vir} > 10^{13} M_{\odot}$

Different selection
criteria:

- Most massive M_{vir}

Protocluster in Magneticum

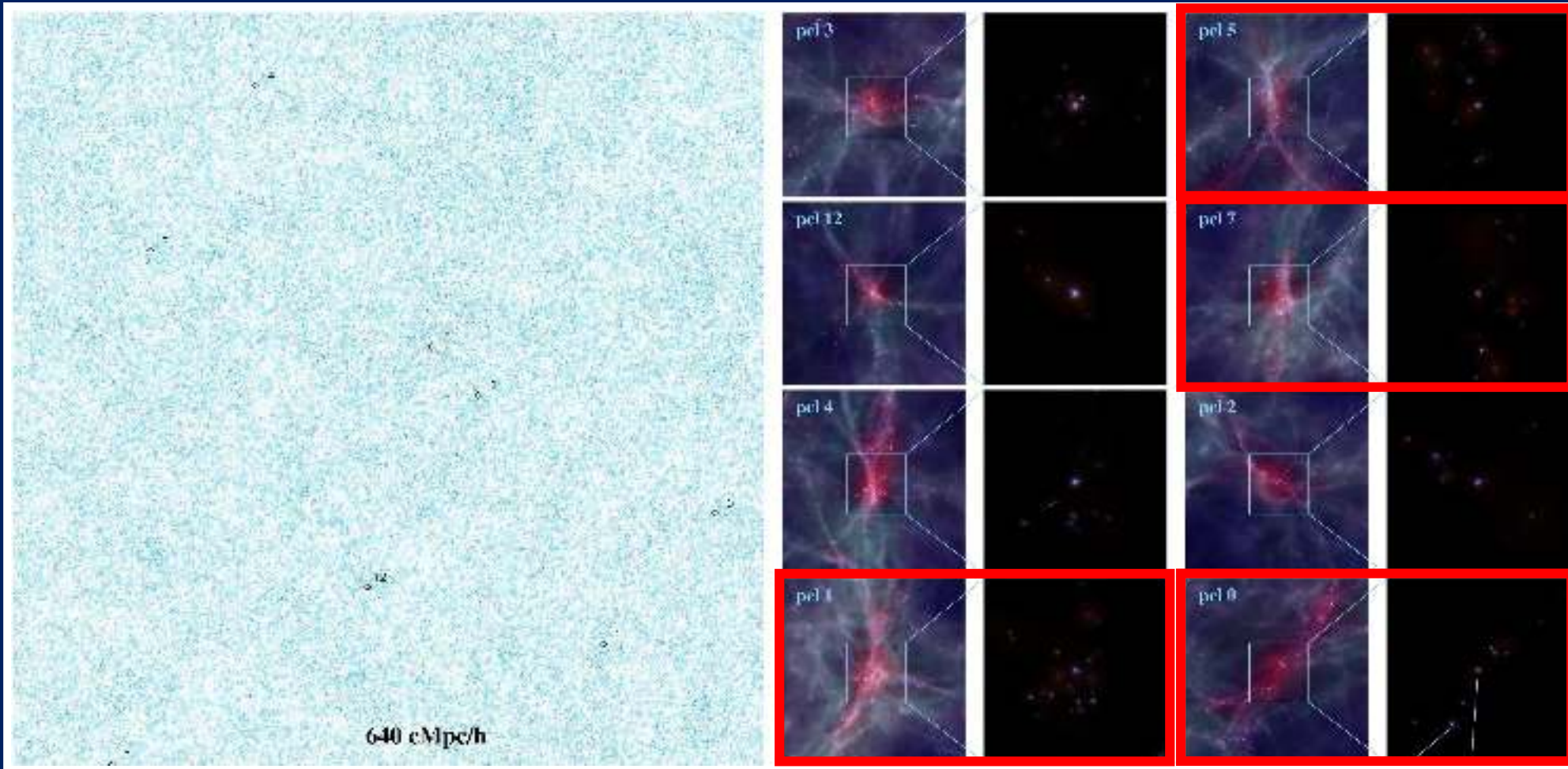


42 potential
Protocluster with
 $M_{vir} > 10^{13} M_{\odot}$

Different selection
criteria:

- Most massive M_{vir}
- Most massive M_{Gal}

Protocluster in Magneticum

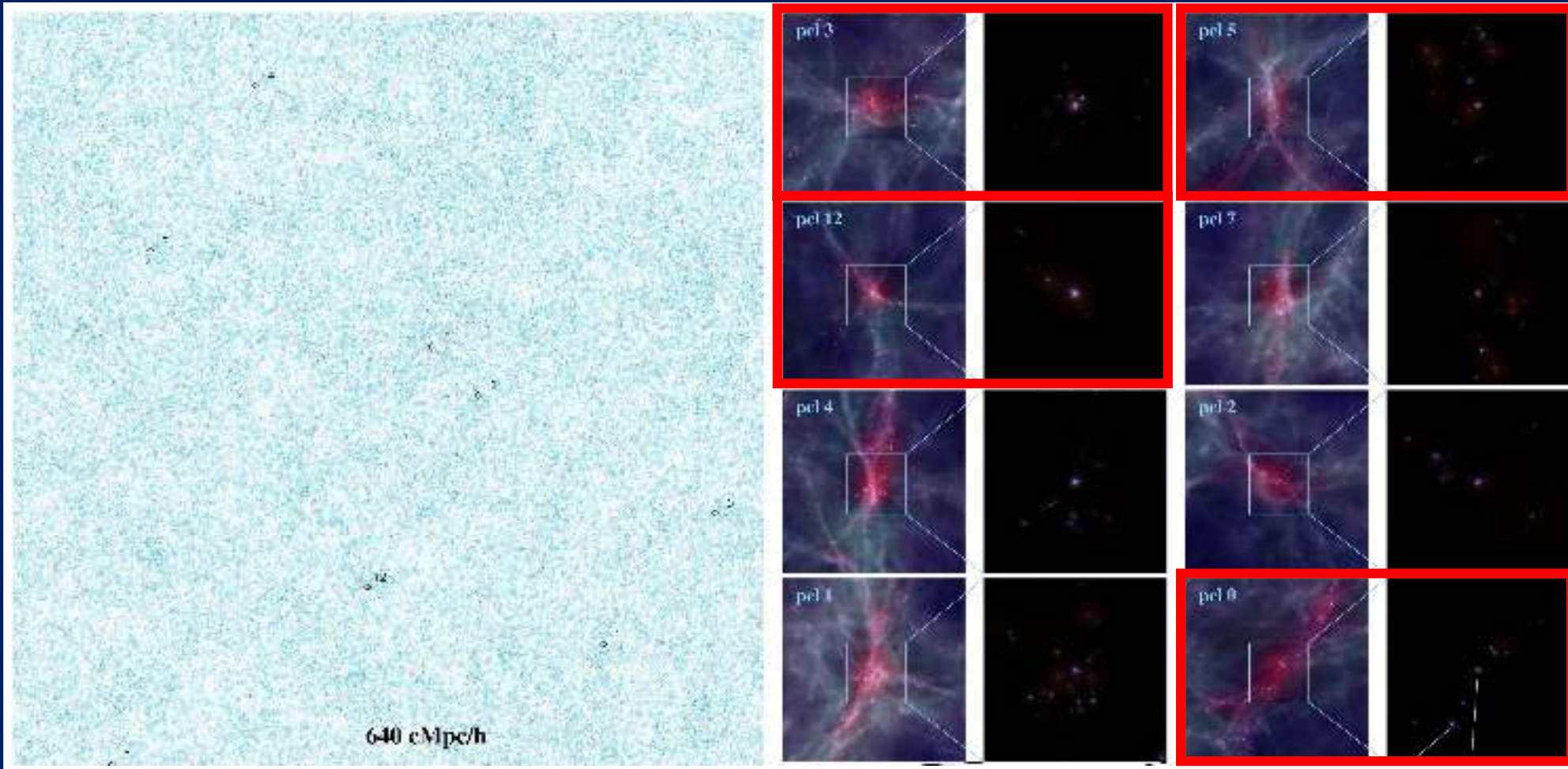


42 potential
Protocluster with
 $M_{vir} > 10^{13} M_{\odot}$

Different selection
criteria:

- Most massive M_{vir}
- Most massive M_{Gal}
- Richest N_{Gal}

Protocluster in Magneticum



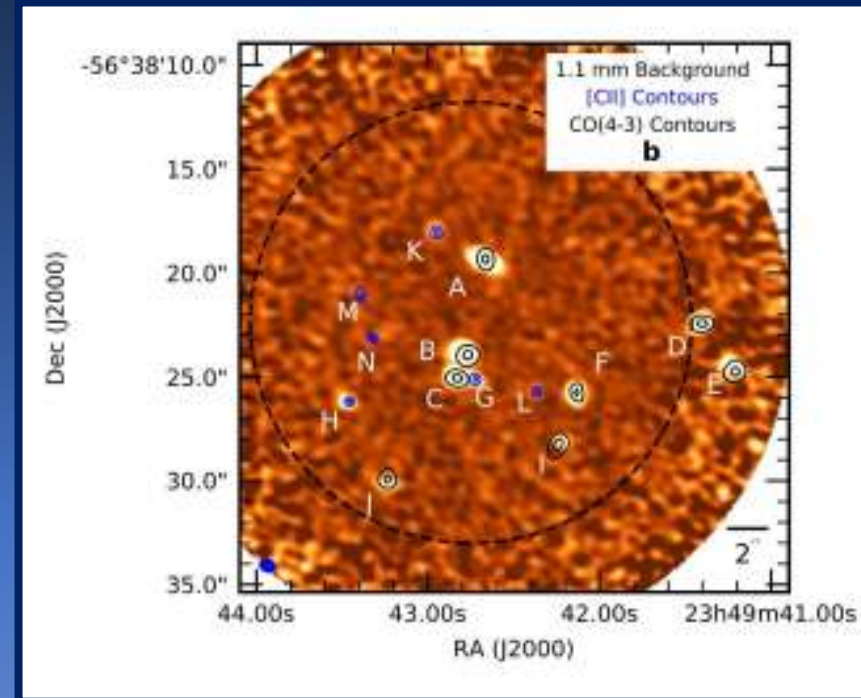
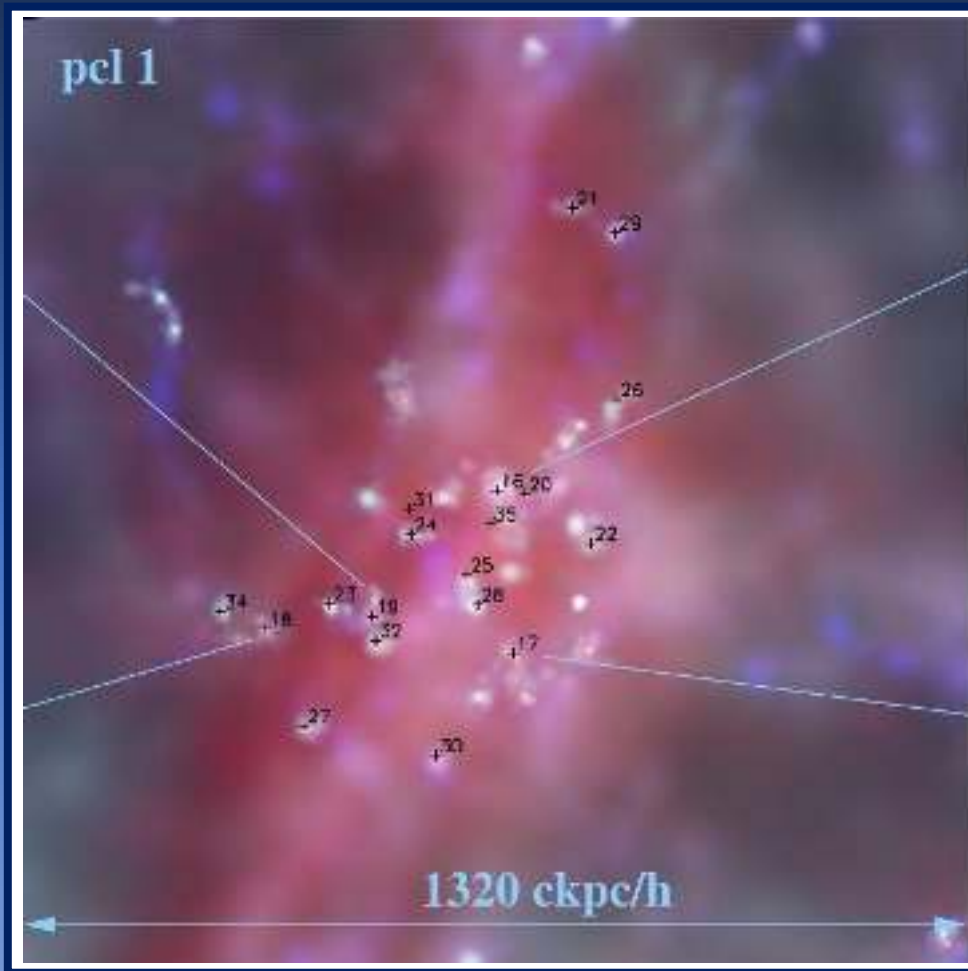
42 potential
Protocluster with
 $M_{vir} > 10^{13} M_{\odot}$

Different selection
criteria:

- Most massive M_{vir}
- Most massive M_{Gal}
- Richest N_{Gal}
- Most star forming

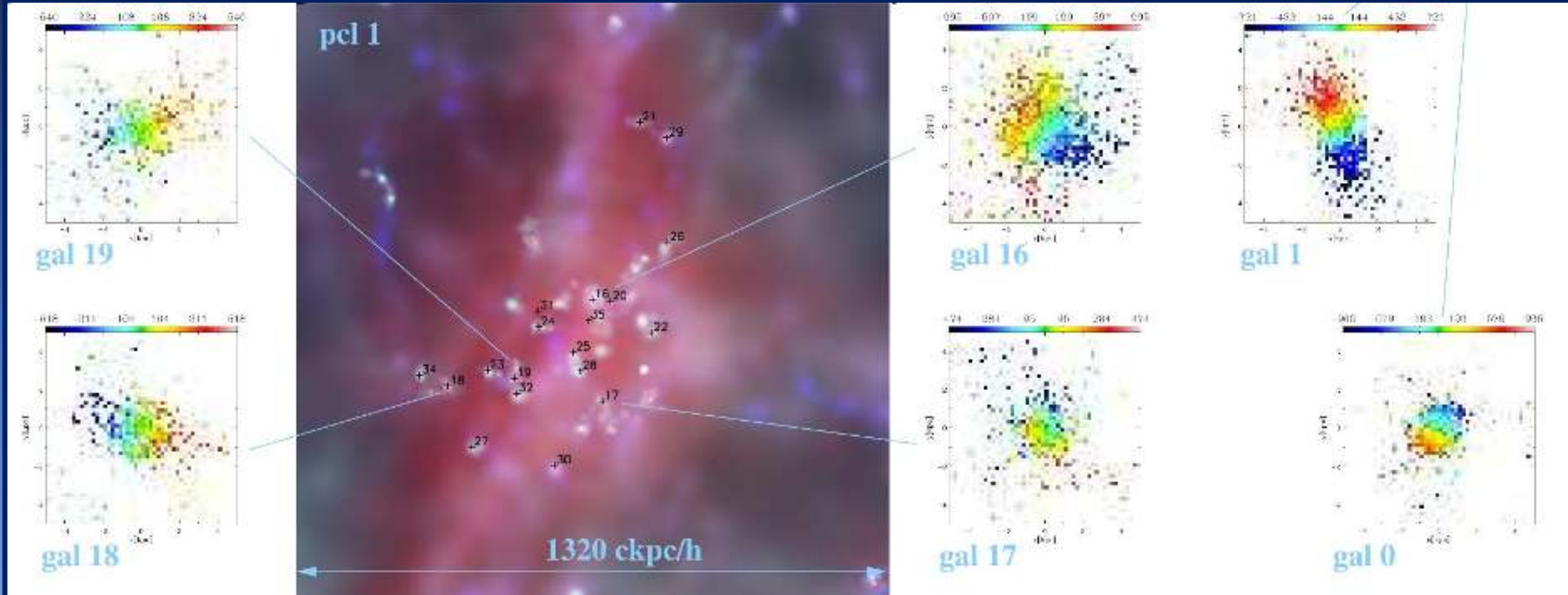


Protocluster in Magneticum



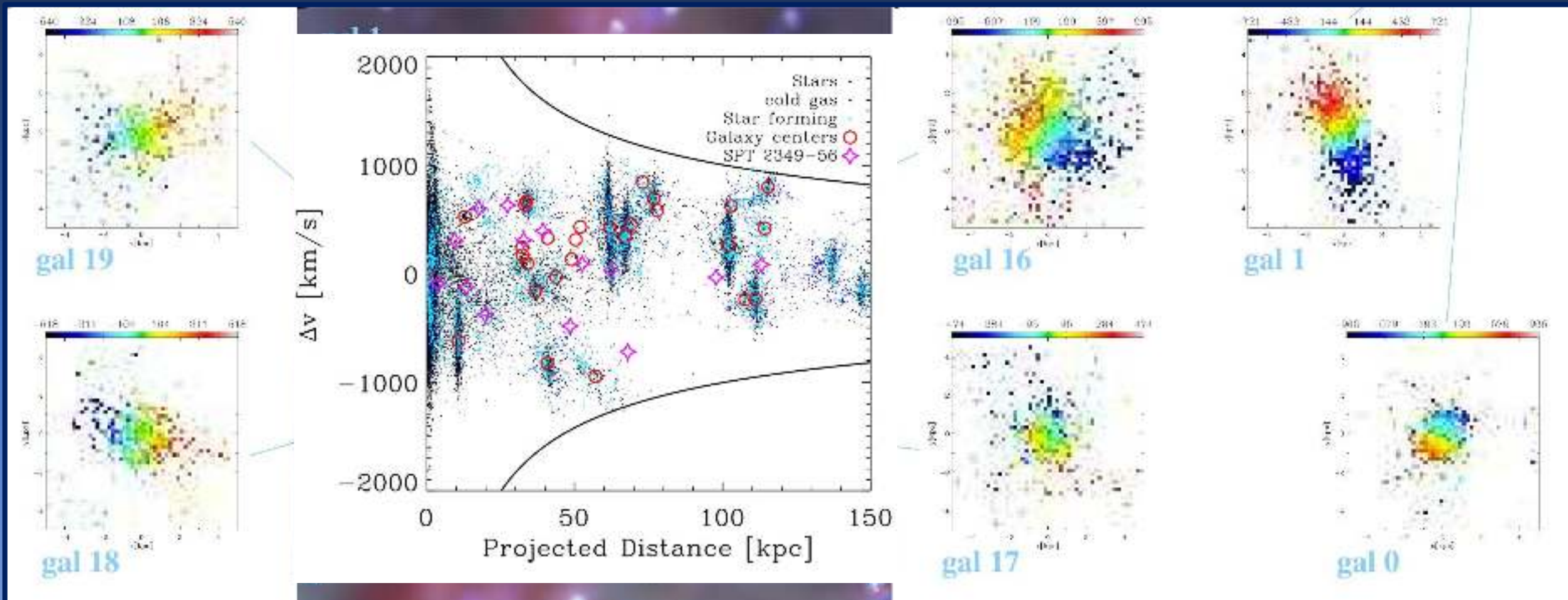
Miller et al., 2018

Protocluster in Magneticum

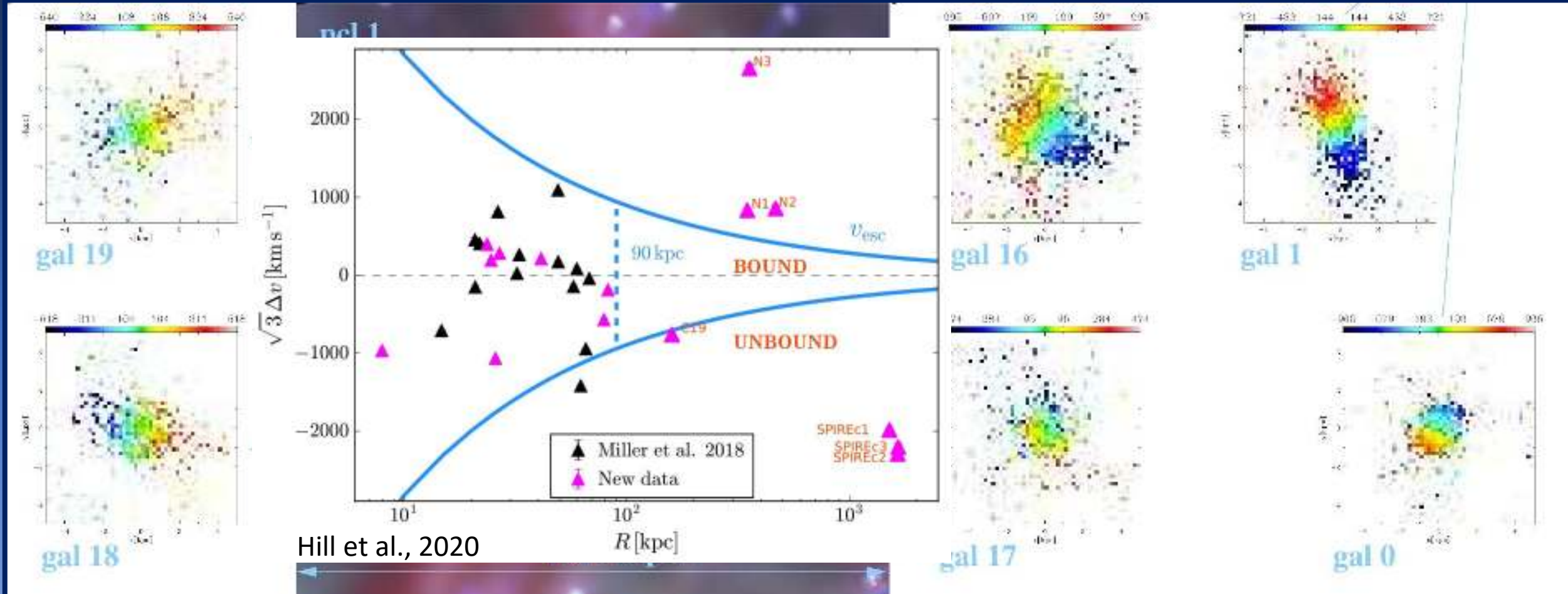




Protocluster in Magneticum

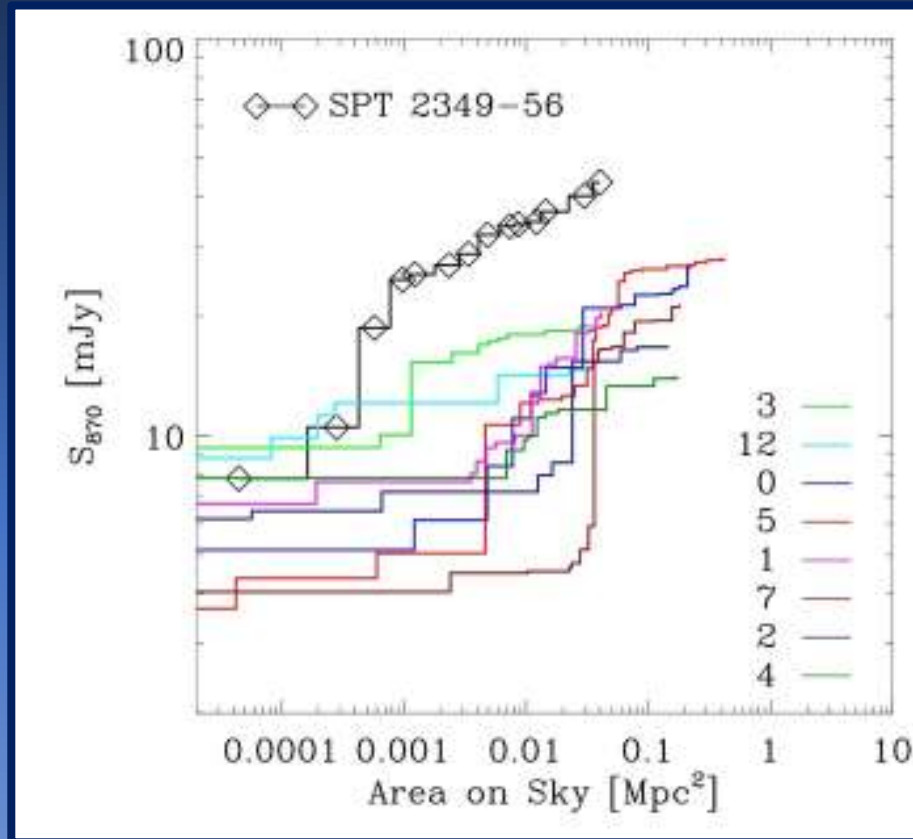
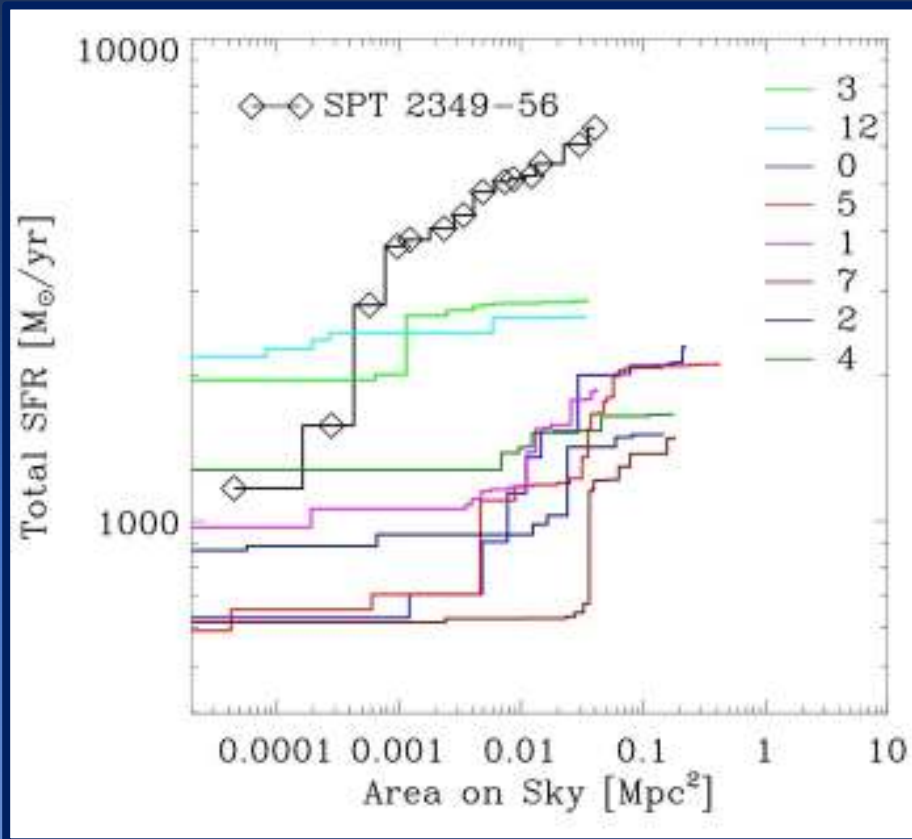


Protocluster in Magneticum





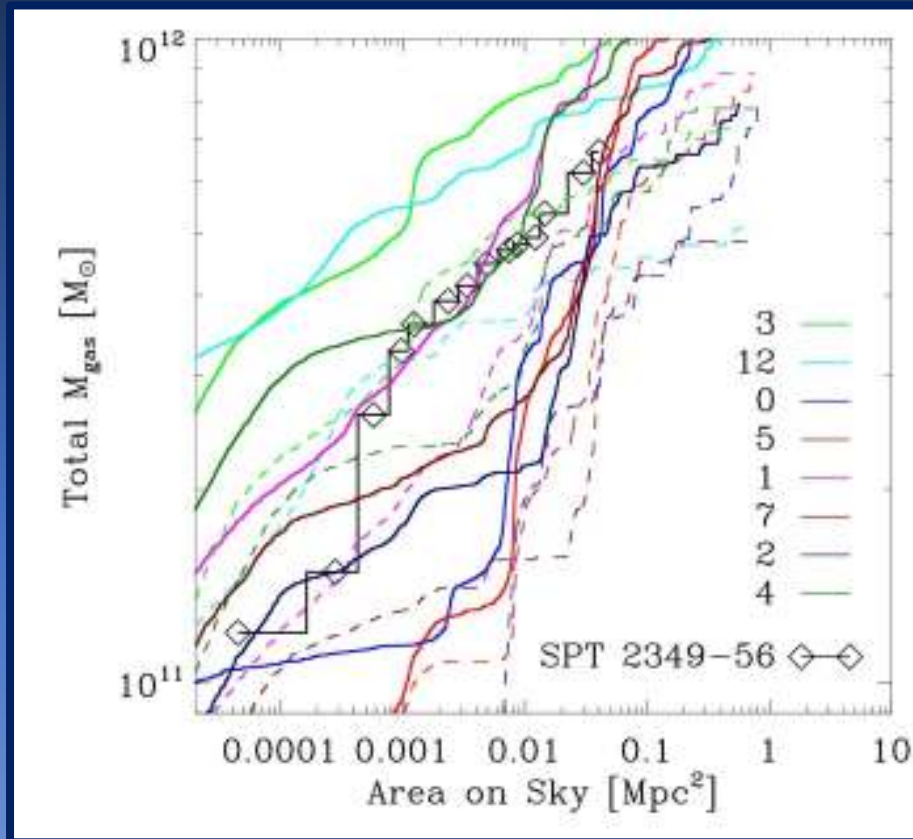
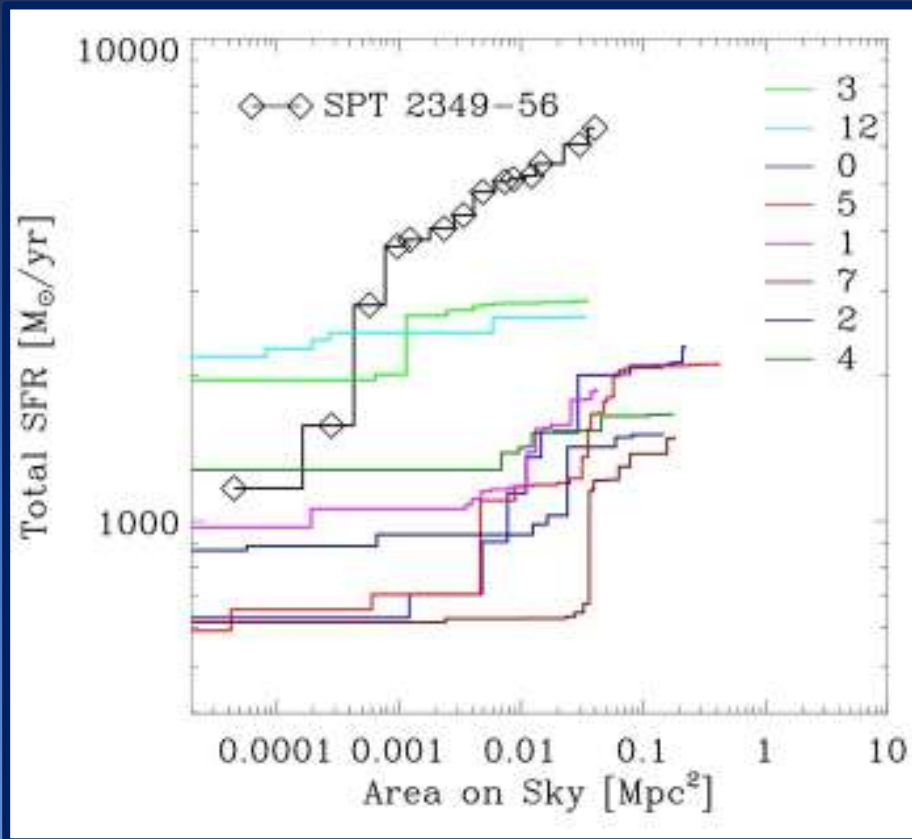
Protocluster in Magneticum: Area on Sky



SPT 2349-56:
Miller et al., 2018

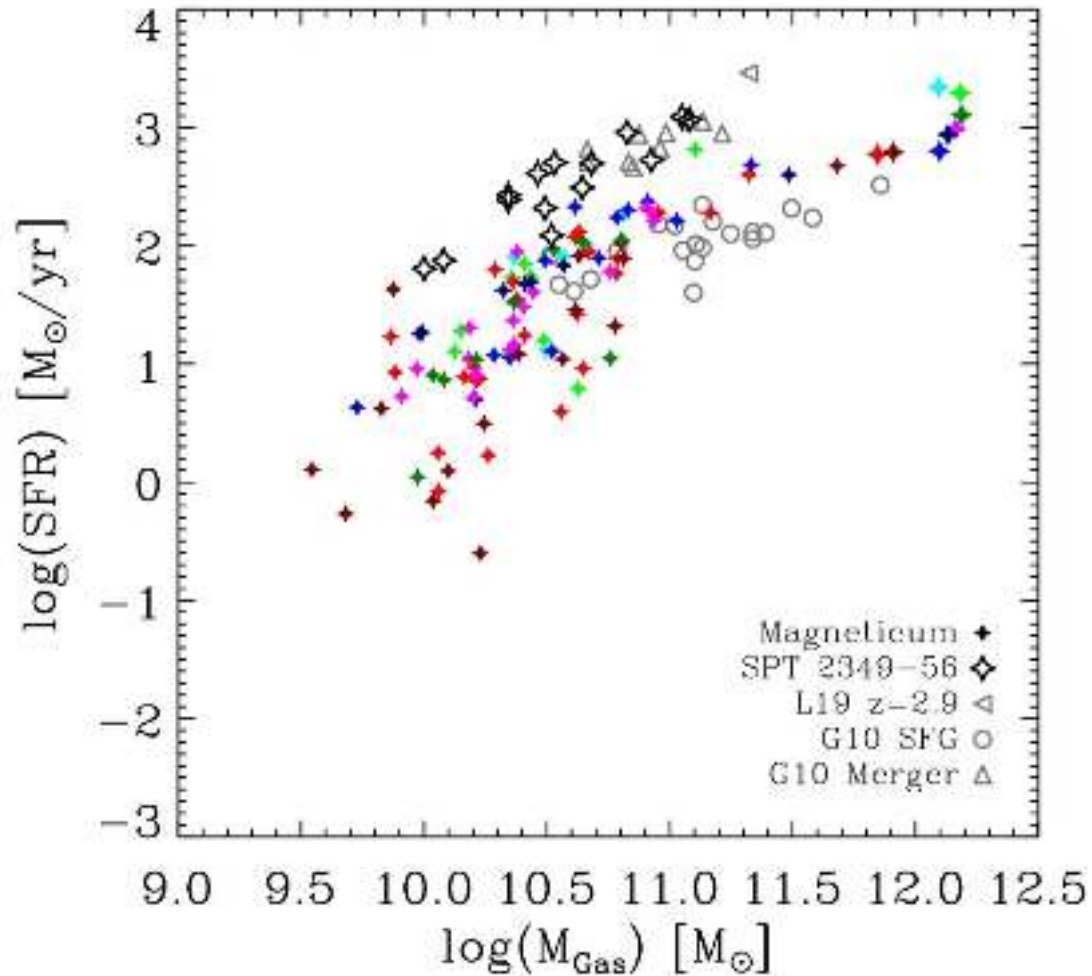


Protocluster in Magneticum: Area on Sky

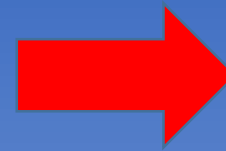


SPT 2349-56:
Miller et al., 2018

Gas mass versus Star Formation Rate



Star formation rate in Magneticum lower than in observed Protocluster galaxies by a factor 2



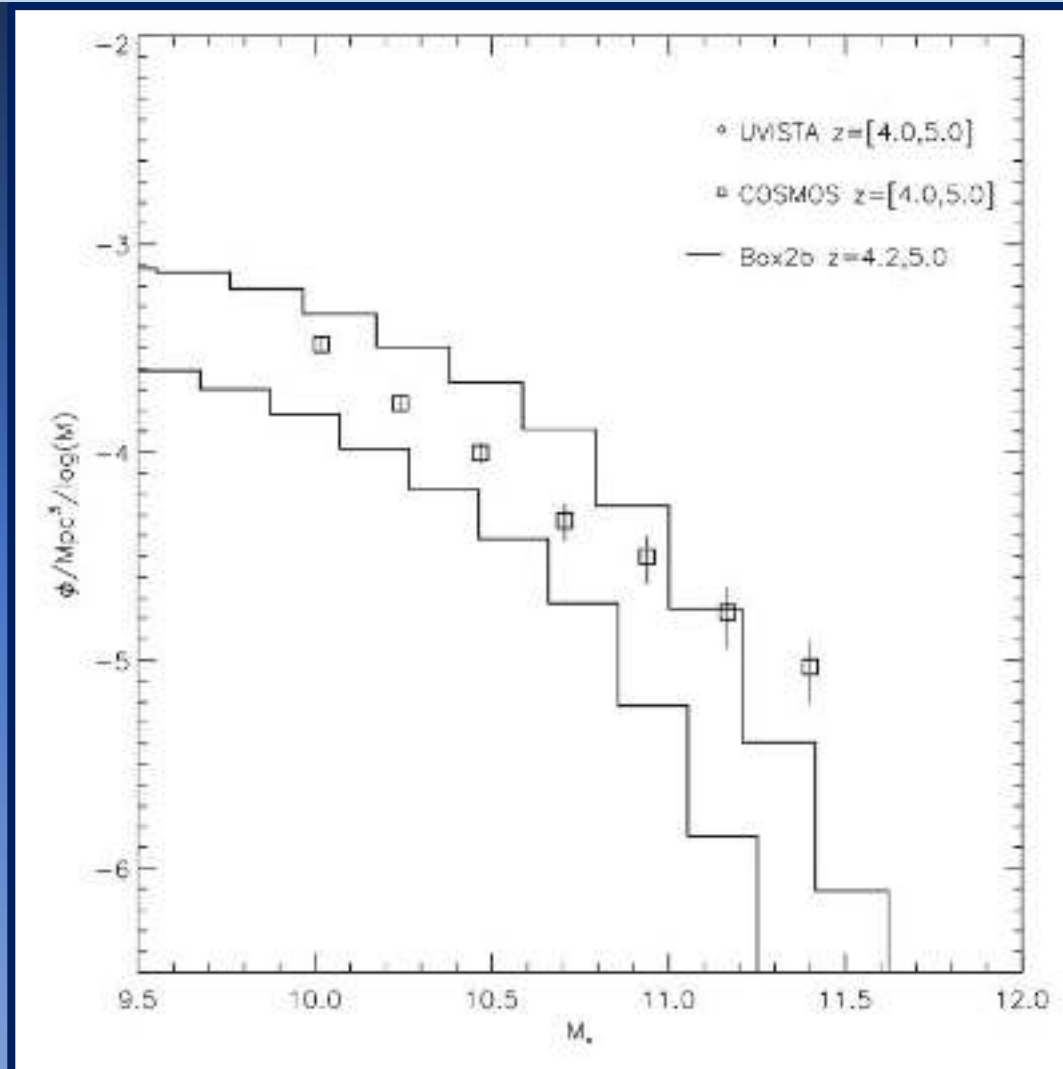
Either star formation at higher redshifts larger, or star formation more bursty than in our model

See also Bassini et al., 2020 for SFR in simulated protoclusters from DIANOGA and his Poster

G10: Genzel et al., 2010: Galaxies with $4 < z < 1$
L19: Leung et al., 2019: HXMM05 @ $z=2.9$

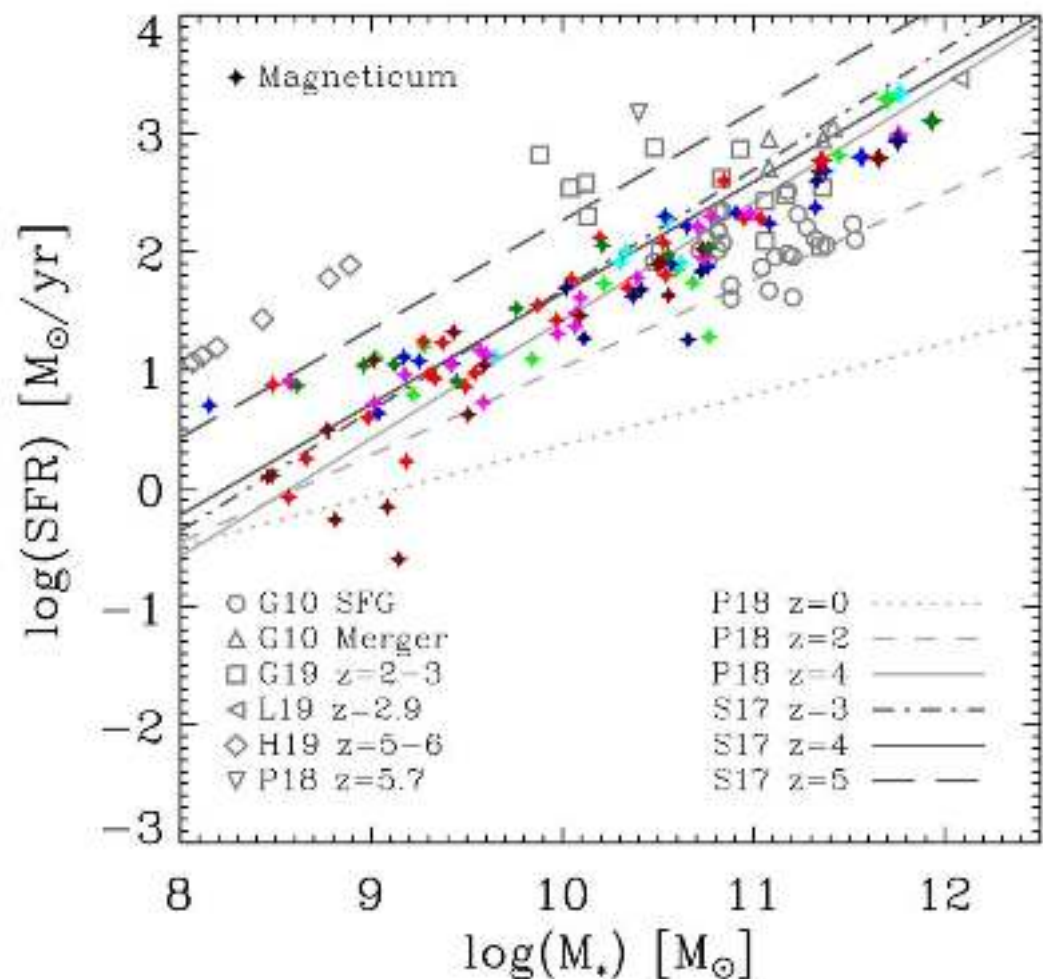


Stellar Mass Function





Star formation main sequence



Star formation main sequence well reproduced by Magneticum even at $z \sim 4$



Star formation more bursty than in our model!

See also Bassini et al., 2020 for SFR in simulated protoclusters from DIANOGA and his Poster

S17: Santini et al., 2017: Hubble frontier field, $6 > z > 1.3$

P18: Pearson et al., 2018: Herschel-SPIRE, $6 > z > 0.2$

G10: Genzel et al., 2010: Galaxies with $4 < z < 1$

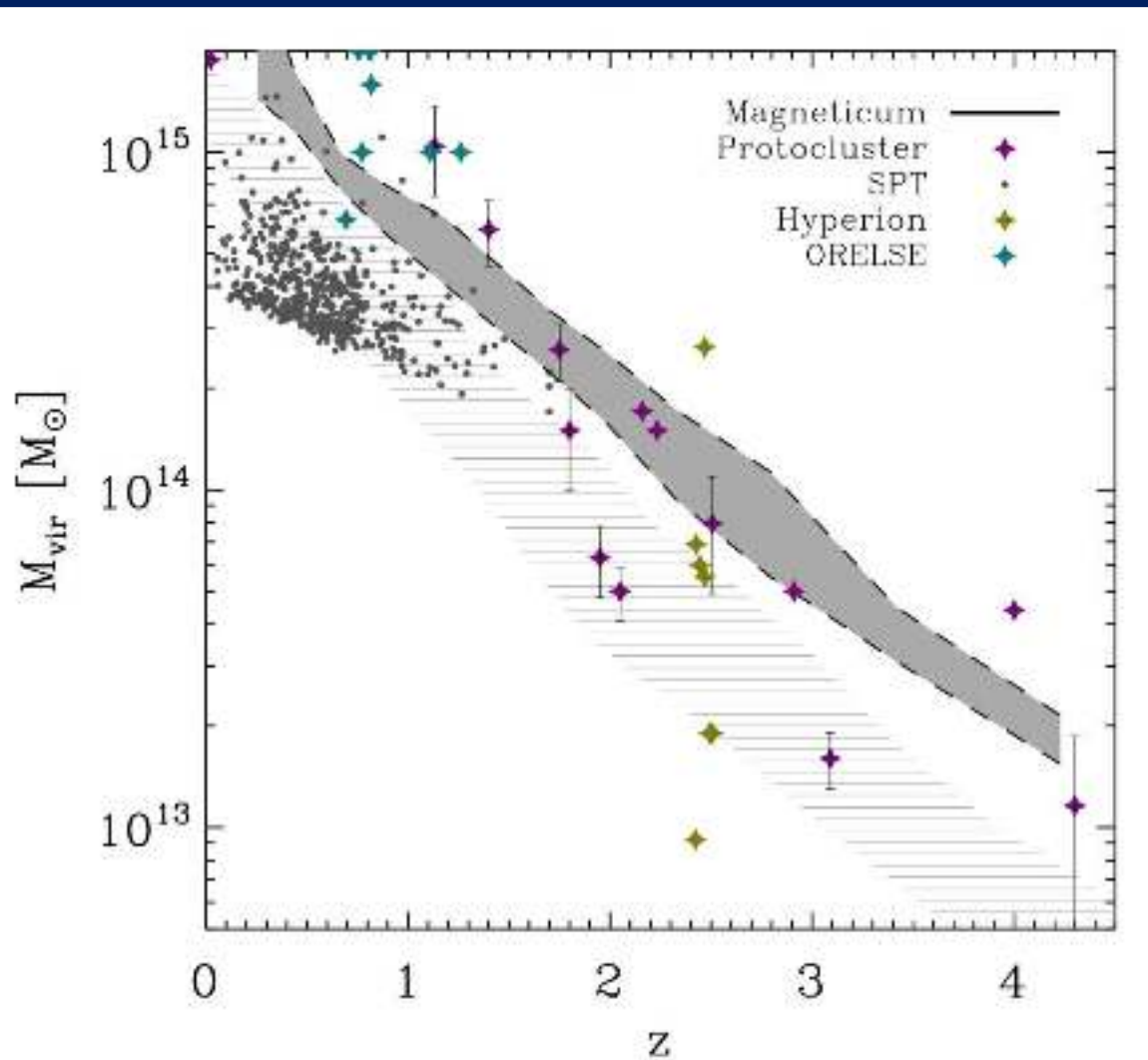
L19: Leung et al., 2019: HXMM05

G19: Gomez-Guijarro et al., 2019: HELAISS02, HXMM20 ($z = 2:6$), CL J1001+0220

P18: Pavesi et al., 2018: CRLE

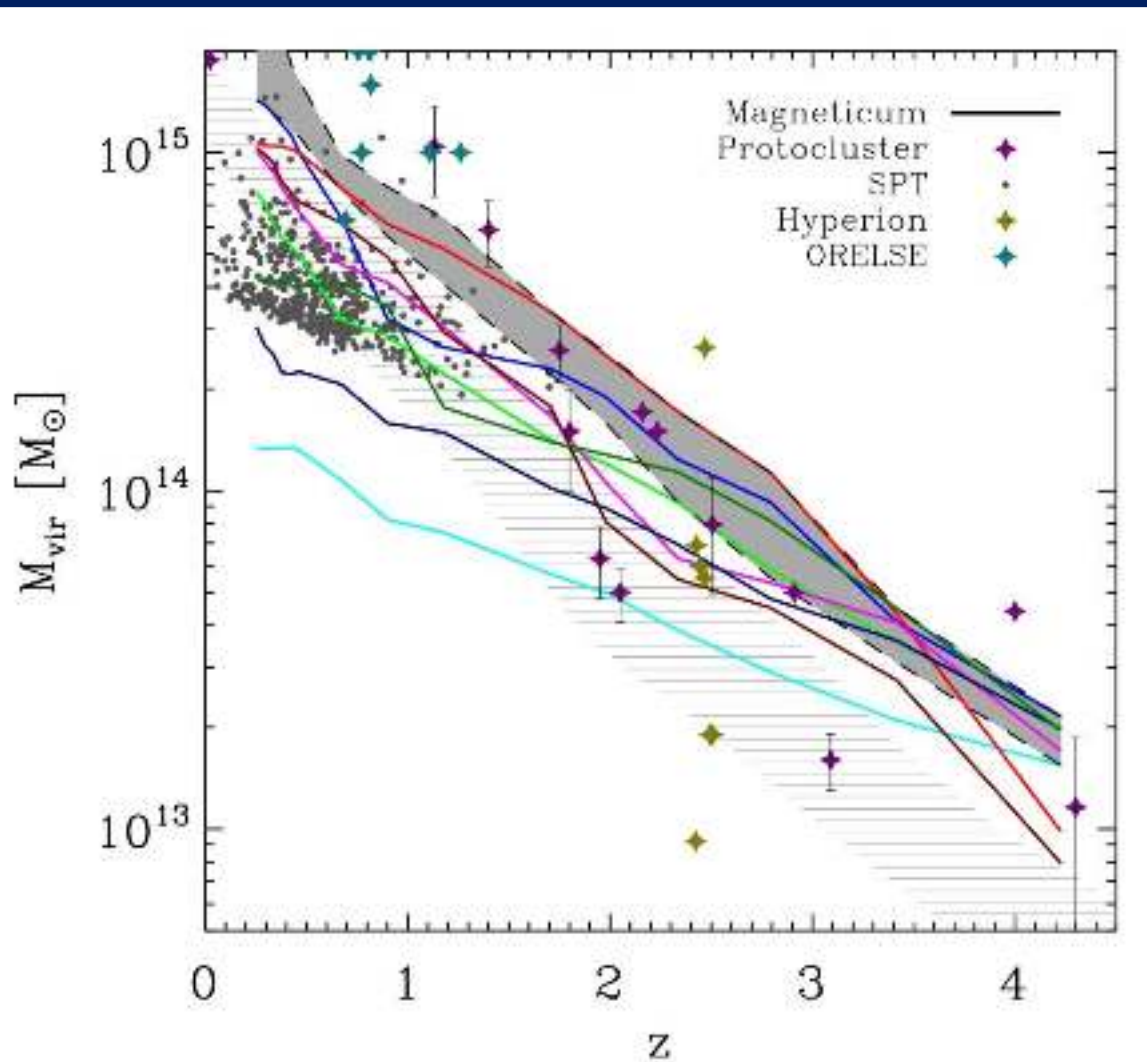
H19: Harikane et al., 2019: Silverrush

Protocluster in Magneticum: Evolution



Shade: Magneticum Most Massive at each redshift
 Striped: Millennium prediction, Chiang et al., 2013

Protocluster in Magneticum: Evolution

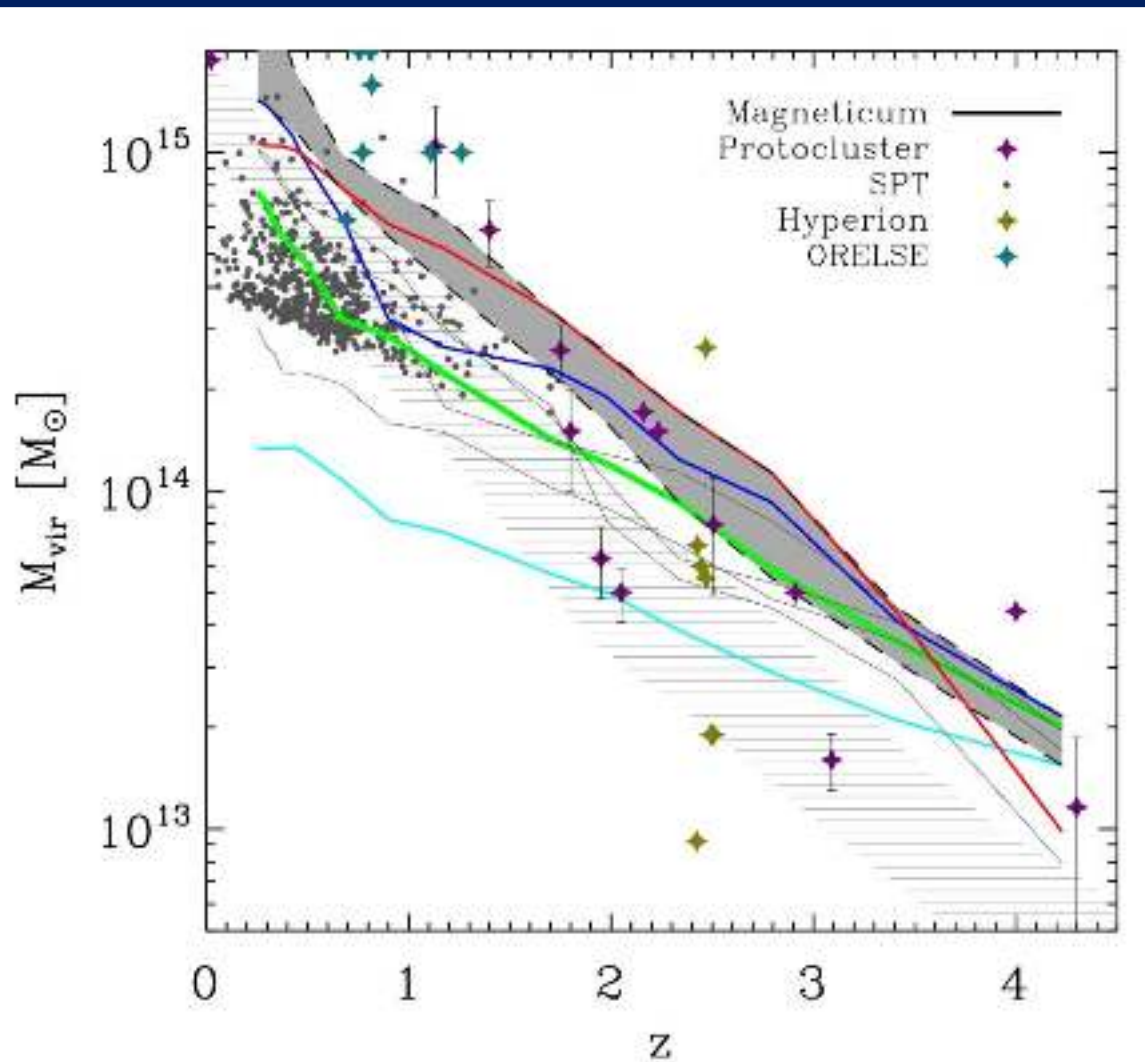


Colored lines: Magneticum Protoclusters
 Shade: Magneticum Most Massive at each redshift
 Striped: Millenium prediction, Chiang et al., 2013

Protocluster in Magneticum: Evolution

Protoclusters: Most Star Forming

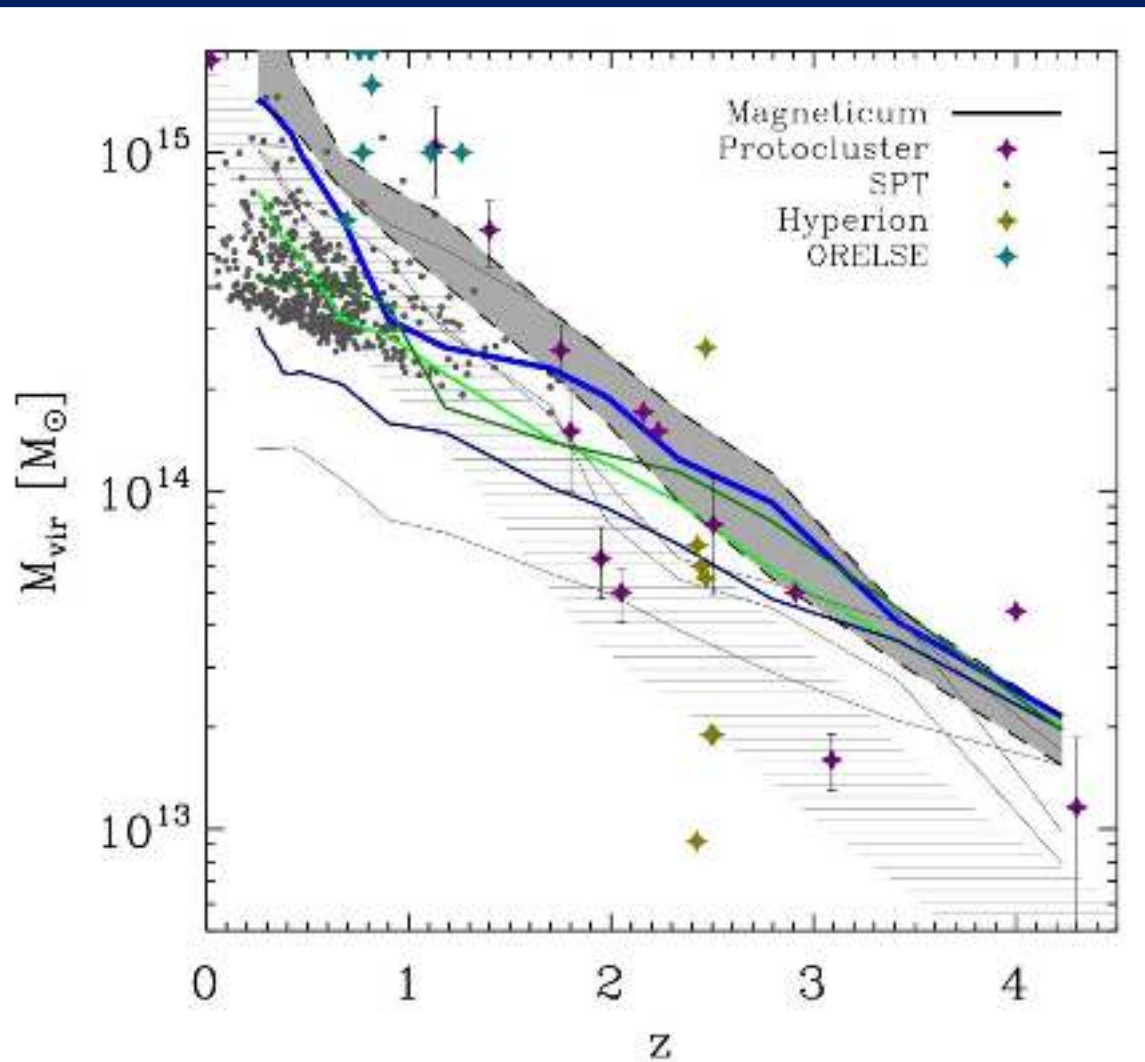
Shade: Magneticum Most Massive at each redshift
Striped: Millenium prediction, Chiang et al., 2013



Protocluster in Magneticum: Evolution

Protoclusters: Most Total Mass

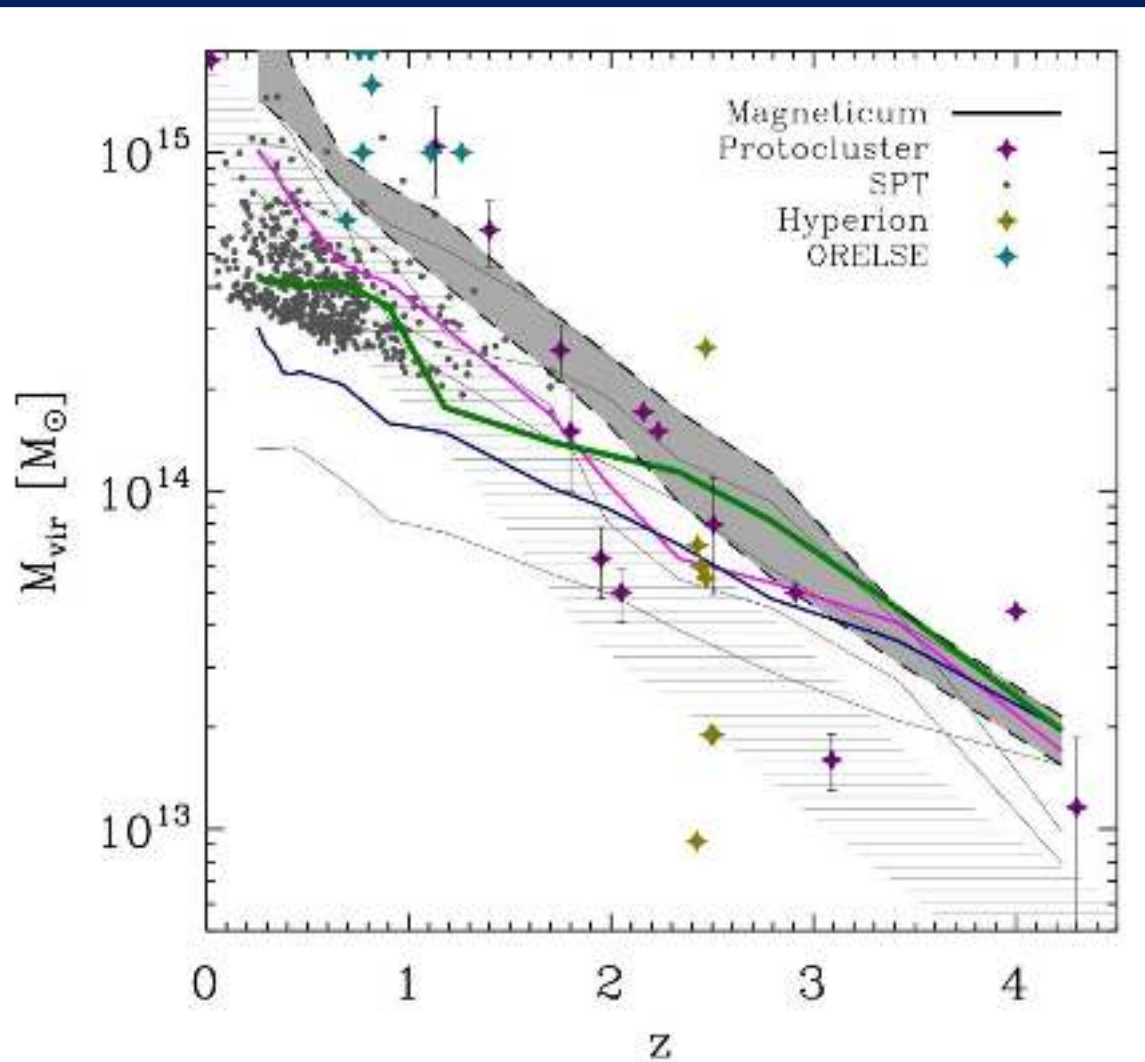
Shade: Magneticum Most Massive at each redshift
Striped: Millenium prediction, Chiang et al., 2013



Protocluster in Magneticum: Evolution

Protoclusters: Most Massive Galaxy

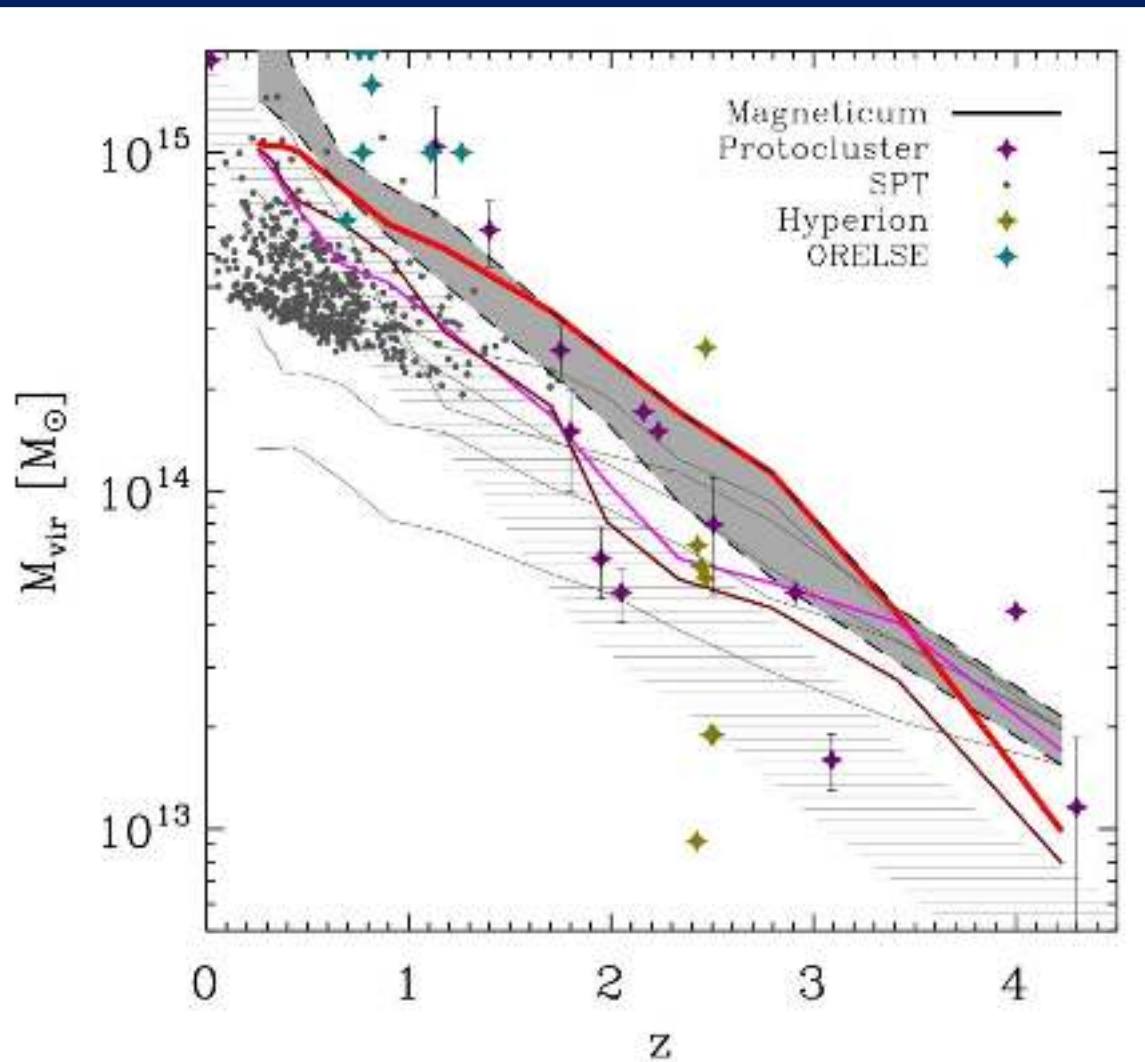
Shade: Magneticum Most Massive at each redshift
Striped: Millenium prediction, Chiang et al., 2013



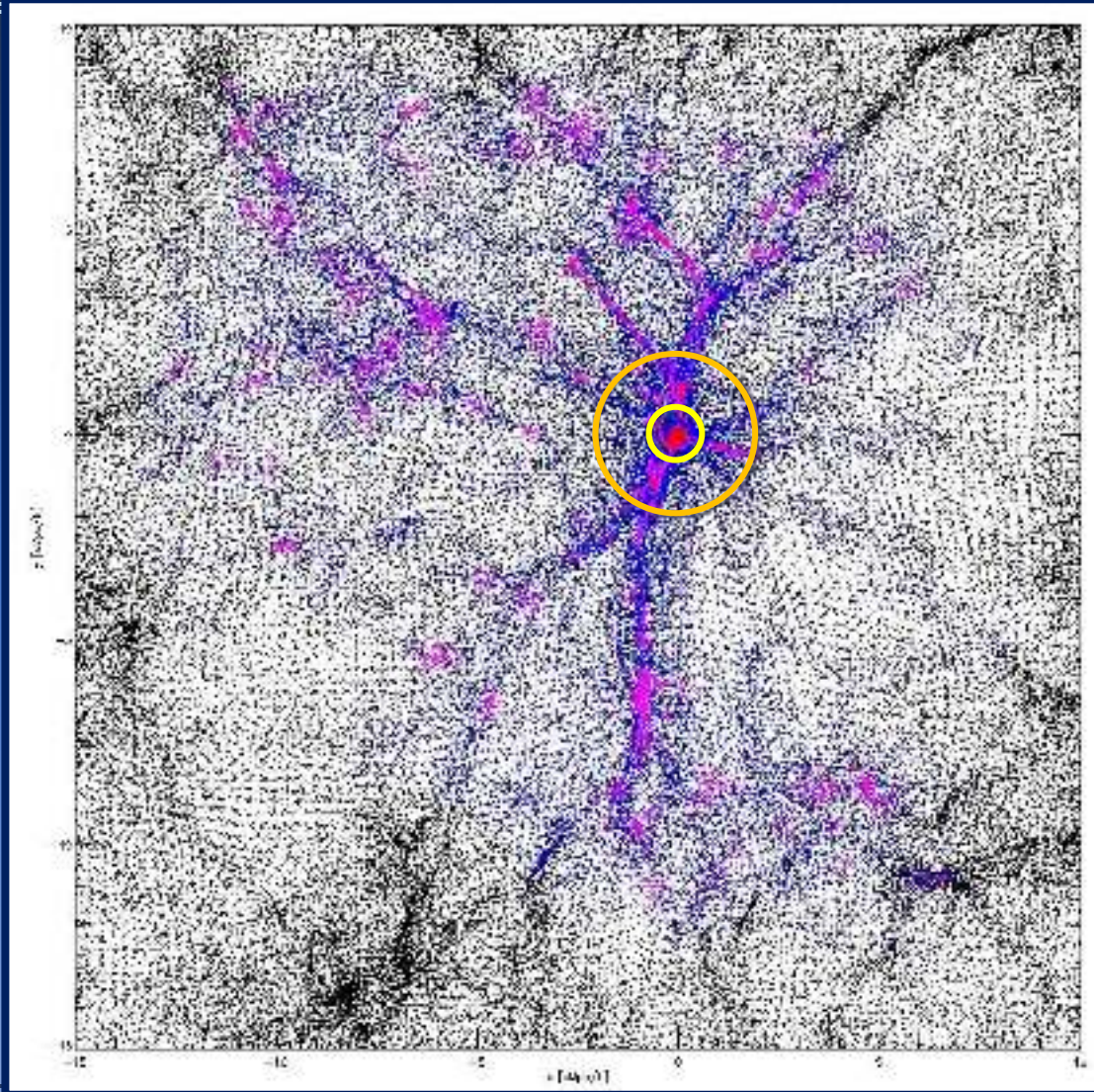
Protocluster in Magneticum: Evolution

Protoclusters: Richest Clusters!

Shade: Magneticum Most Massive at each redshift
Striped: Millenium prediction, Chiang et al., 2013



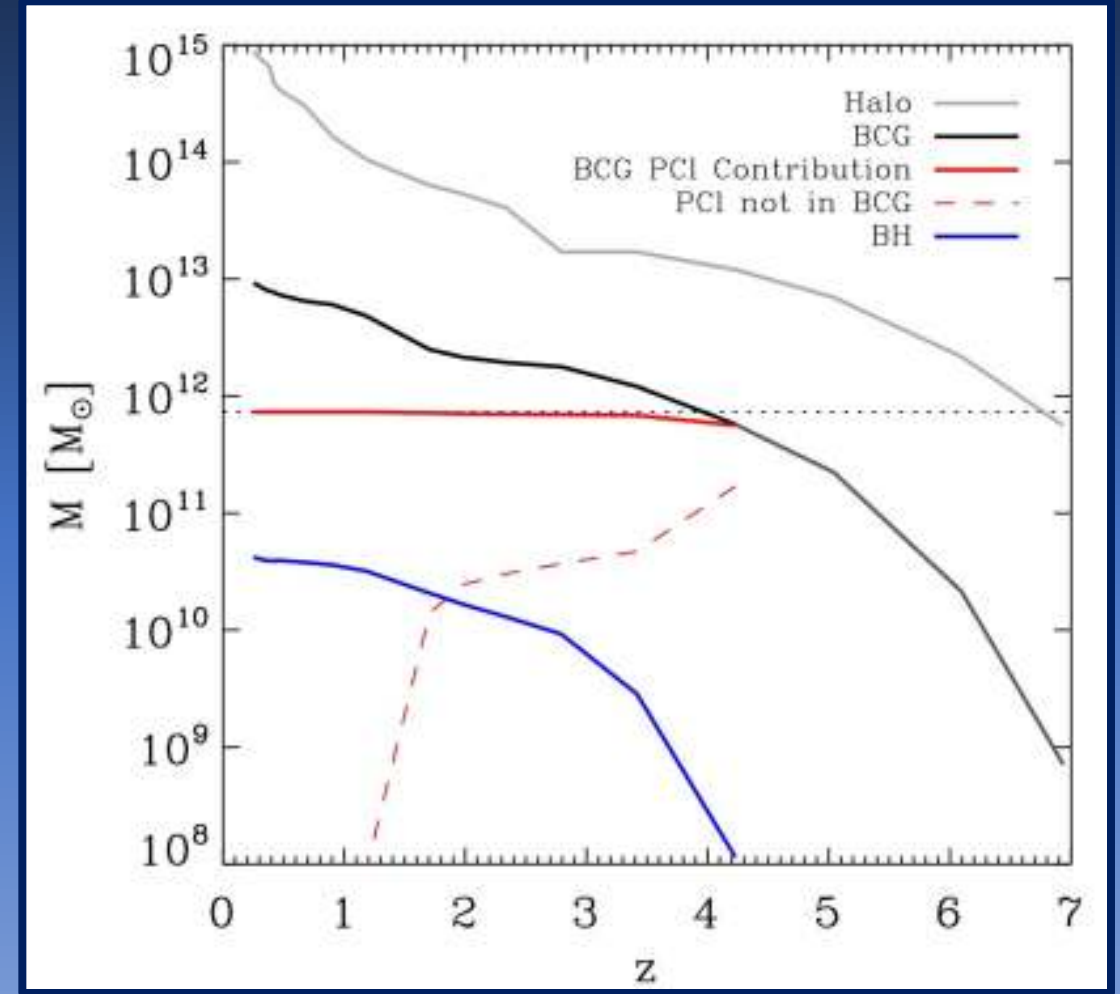
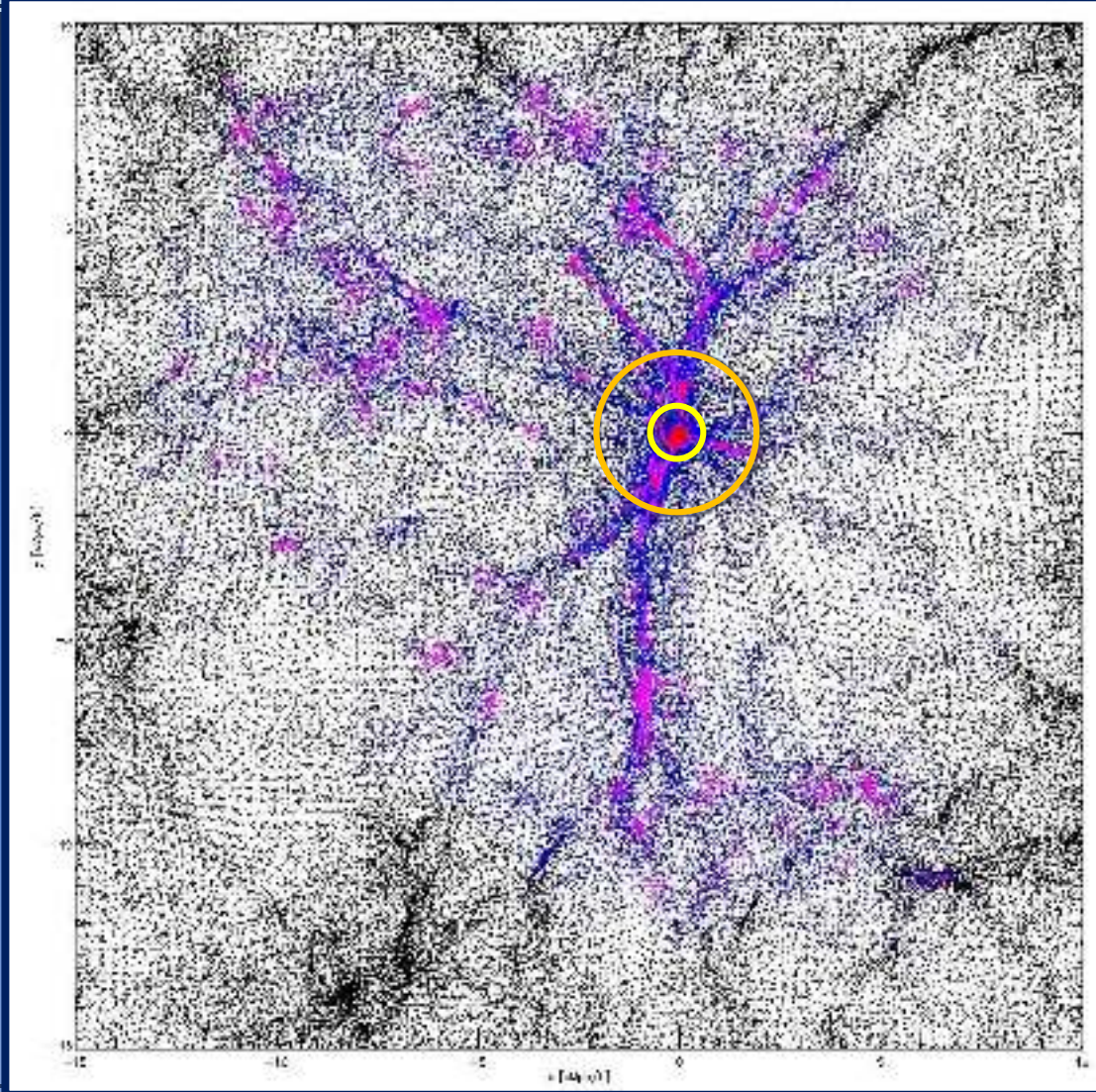
Protocluster in Magneticum: Evolution



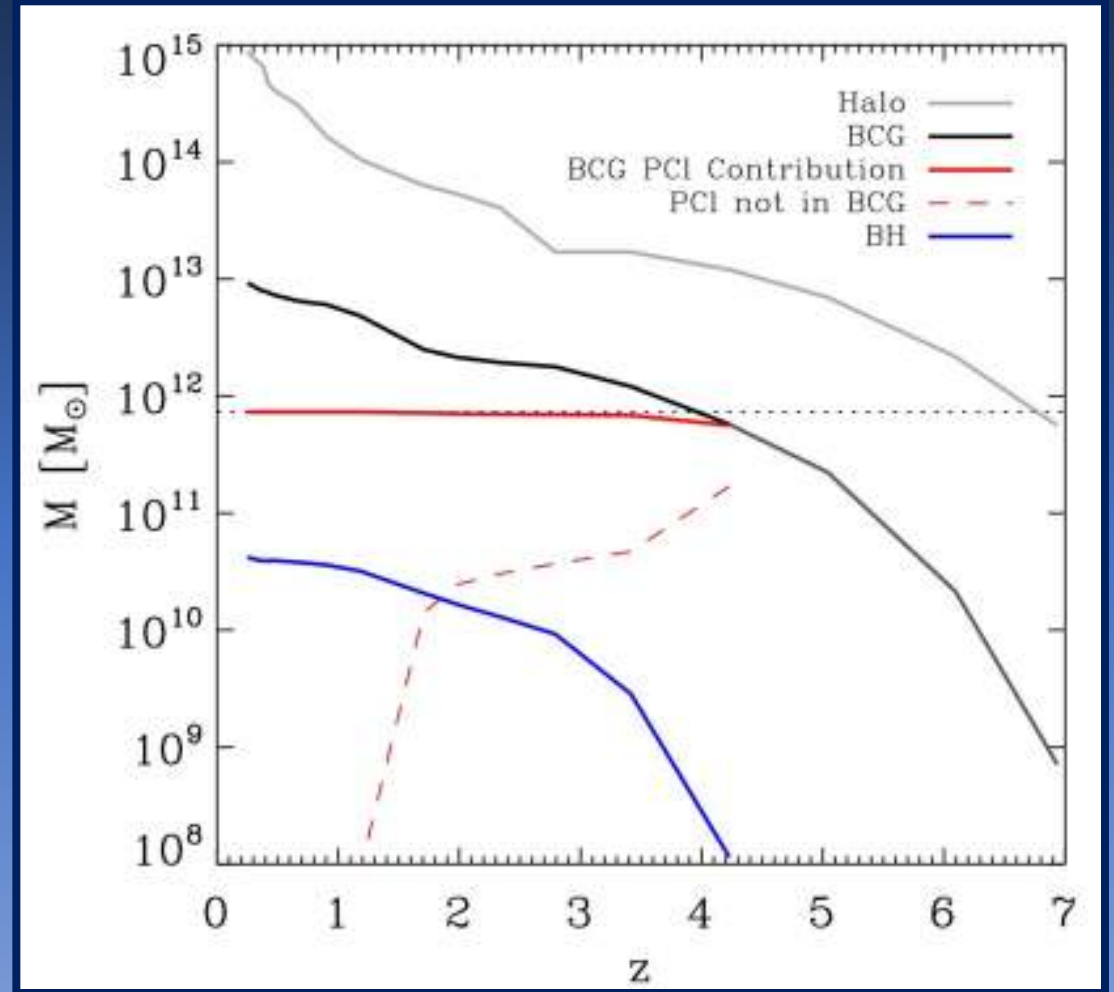
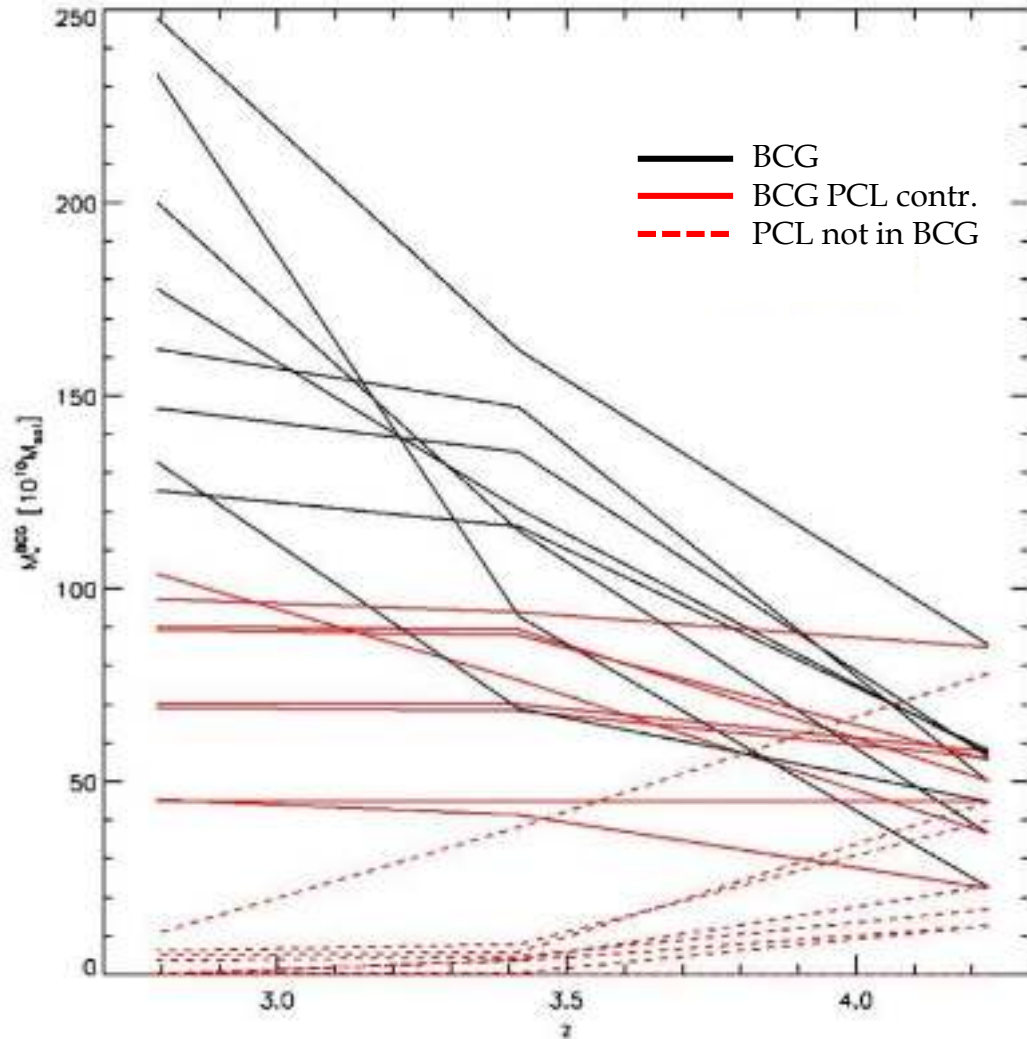
$R_{vir} @ z = 4.3$

$R_{vir} @ z = 0$

Protocluster in Magneticum: BCG Assembly

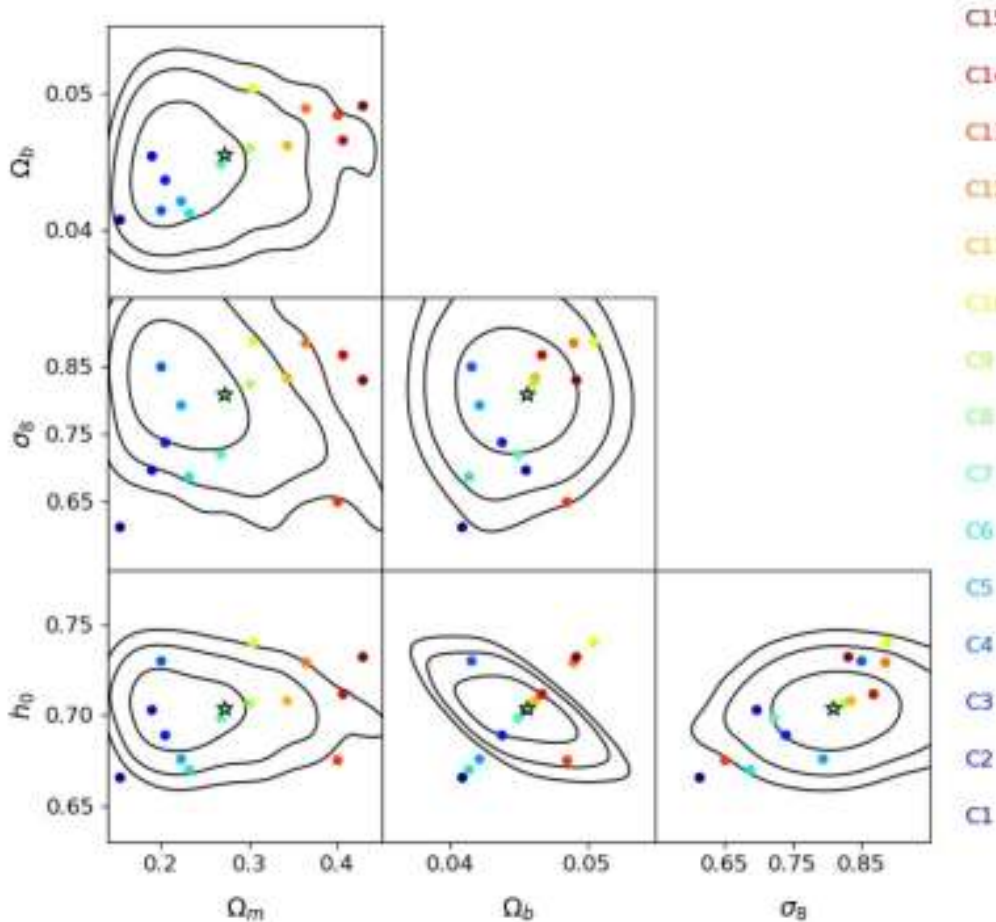


Protocluster in Magneticum: BCG Assembly



See also Rennehan et al., 2019 for BCG formation in Simulations and his Poster

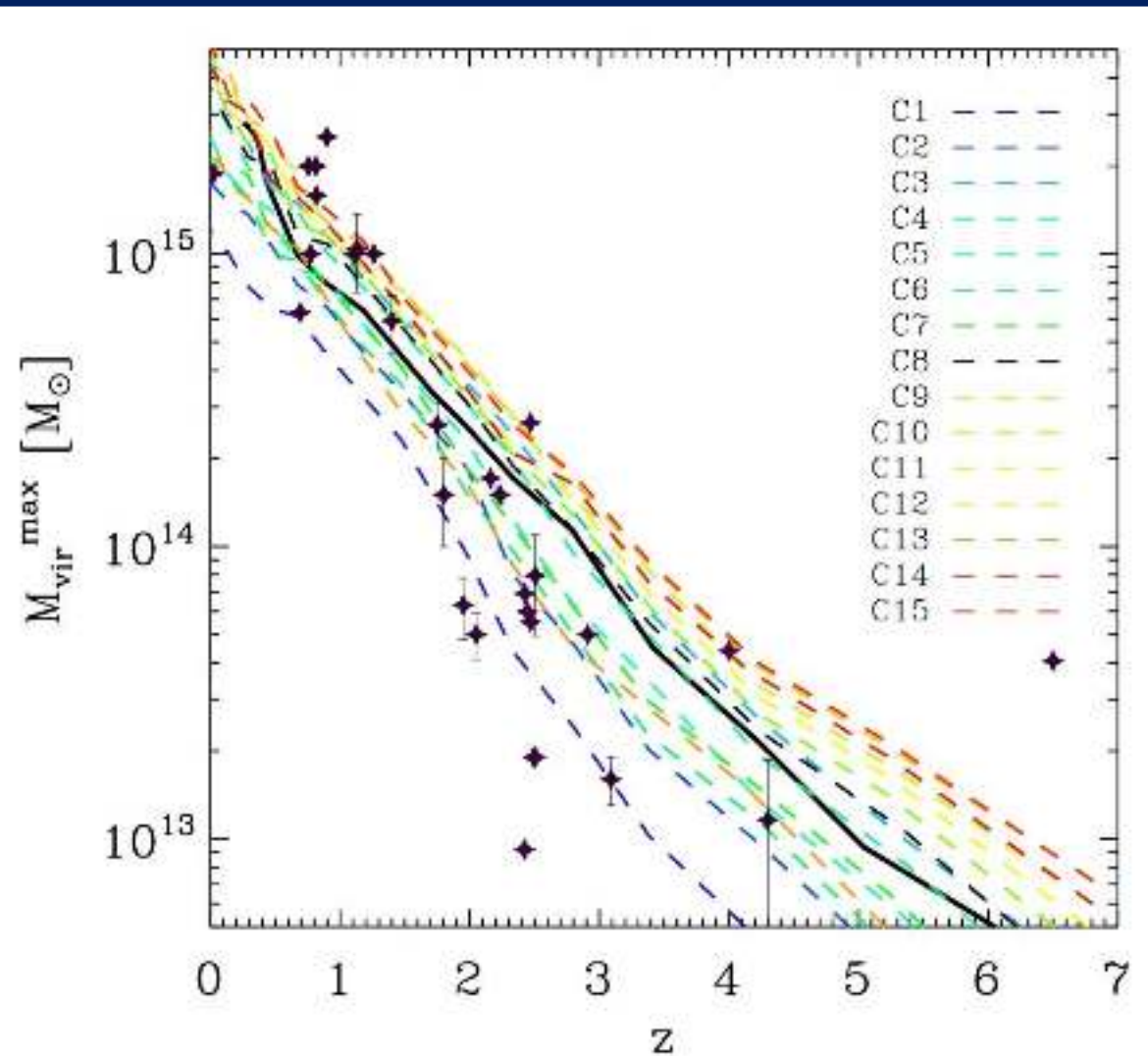
Protocluster in Magneticum: Cosmology Test



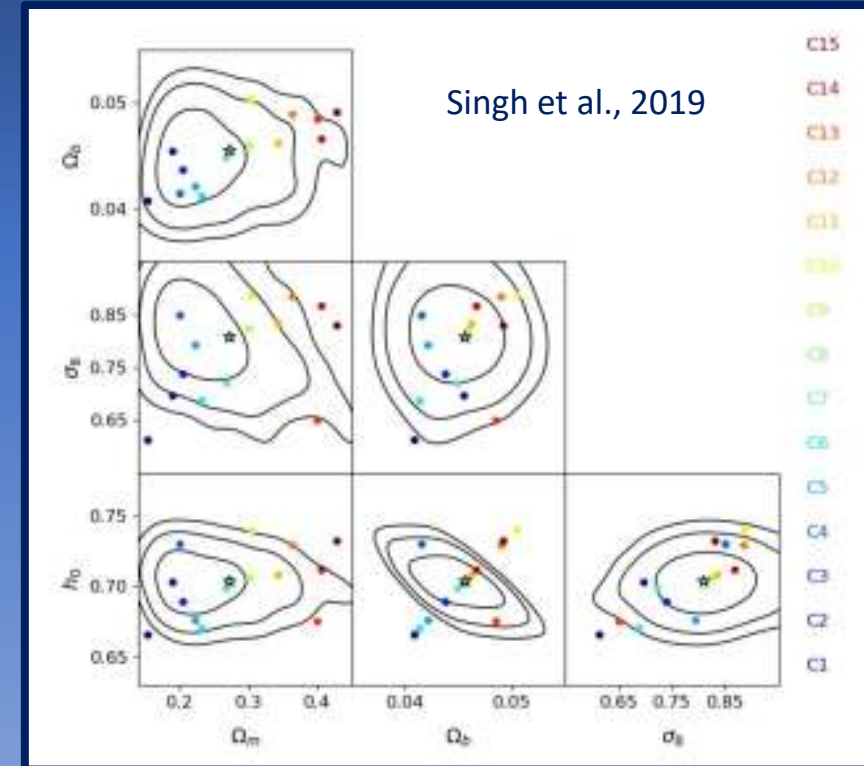
	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15
Ω_m	0.153	0.189	0.200	0.204	0.222	0.232	0.268	0.272	0.301	0.304	0.342	0.363	0.400	0.406	0.428
Ω_b	0.0408	0.0455	0.0415	0.0437	0.0421	0.413	0.0449	0.0456	0.0460	0.0504	0.0462	0.0490	0.0485	0.0466	0.0492
σ_8	0.614	0.697	0.850	0.739	0.793	0.687	0.721	0.809	0.824	0.886	0.834	0.884	0.650	0.867	0.830
h_0	0.666	0.703	0.730	0.689	0.676	0.670	0.699	0.704	0.707	0.740	0.708	0.729	0.675	0.712	0.732
f_h	0.267	0.241	0.208	0.214	0.190	0.178	0.168	0.168	0.153	0.166	0.135	0.135	0.121	0.115	0.115

Singh et al., 2019. The star marks the WMAP-7 cosmology used for the standard Magneticum Pathfinder runs

Protocluster in Magneticum: Cosmology Test



σ_8 most important for slope of $M_{\text{vir}}-z$ relation.



Summary & Conclusions

- Massive protoclusters like SPT2349-56 can be found in Magneticum at $z \sim 4.3$.
- Number of member galaxies, phase-space properties, spatial distributions, and cold gas properties are well captured by the simulation.
- Star formation rates are too low, but gas mass and star formation main sequence are well captured even at high redshifts.
 - ➔ Star formation more bursty than included in our model!
- Our protoclusters at $z = 4.3$ evolve into halo at $z = 0$ covering a large range of masses.
 - ➔ Not all of them are massive clusters at $z=0$!
- Neither mass nor star formation at $z=4.3$ are a good measure of what to become, but the number of galaxies is – cosmic web tracing!
- BCG core assembles quickly from these protocluster-structures (about 1Gyr)
- Can be used to test cosmology

www.magneticum.org

